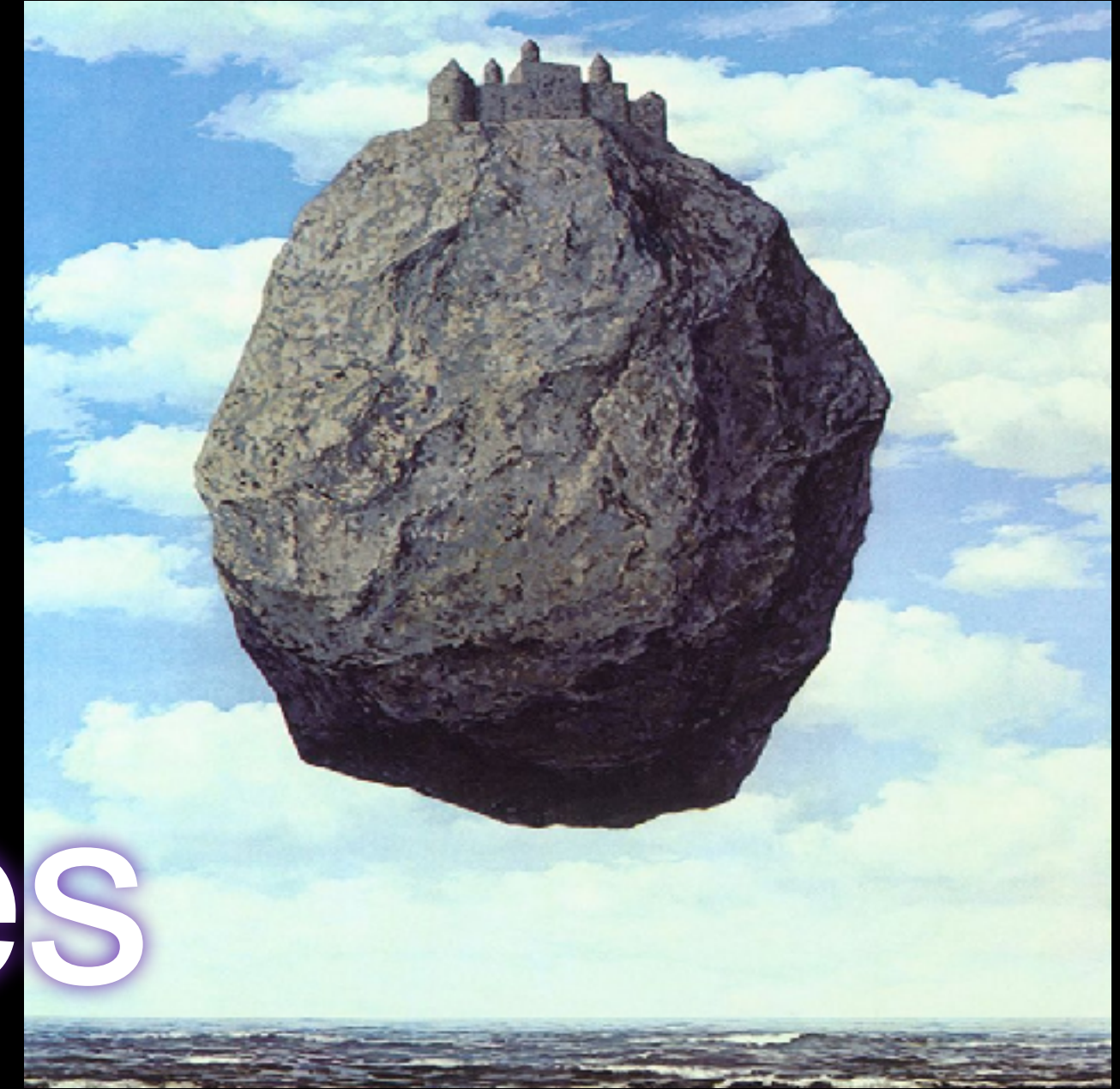


Superposition

HEAD Master Media Design

Invisible Cities



November 2025

Invisible Cities

Day 1

Briefing

Invisible Cities

In this workshop...

Day 1

You will explore the possibilities of procedural design in this workshop inspired by Italo Calvino's "Invisible Cities".

Over the course of one week, you will experiment with Blender Geometry Nodes to design your own interpretation of a fictional cities.

You will work in groups composed of first-year and second-year students, with roles that focus loosely on architectural and natural elements within a procedural city that you build together.

On the final day, you will present your city in the physical space, for instance, by projecting onto an object or treating a screen as a physical representation of your city in 3d space.

Schedule

Day 1

On **Monday**, you will get inspired, learn the basics of Geometry Nodes, and design a concept for your city and its physical installation.

On **Tuesday**, you will learn the core techniques for building both natural and architectural systems in two hands-on masterclasses.

On **Wednesday**, you will learn how to make your two systems interact and merge them into one unified, generative city.

On **Thursday**, you will learn to translate your digital world into a physical one by exporting models, preparing renders, and starting fabrication.

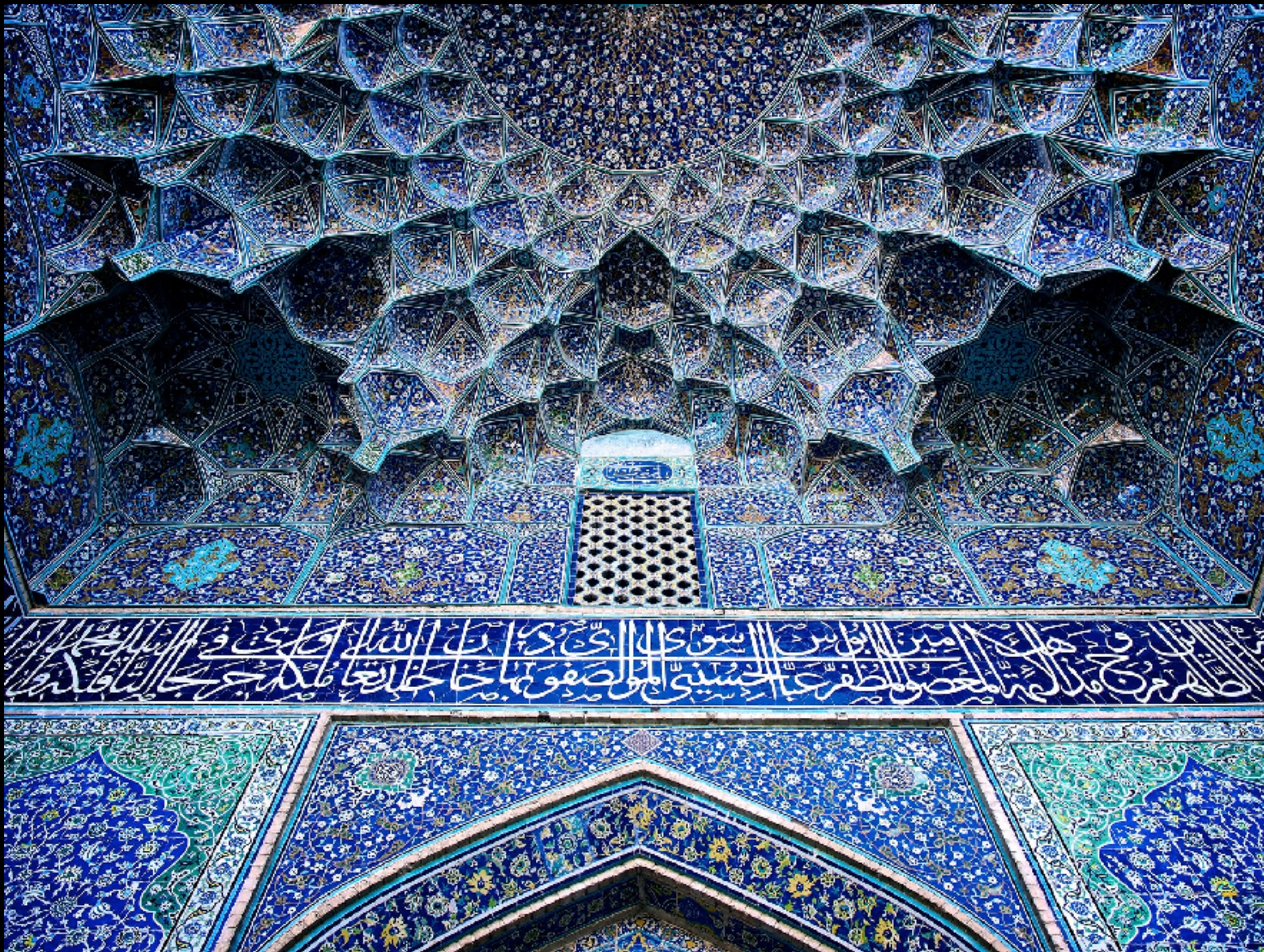
On **Friday**, you will install your physical and digital work in the space and present it to the group in a final gallery walk exhibition.

Inspiration

Muqarnas

3D geometric vaults found in Islamic and Persian architecture (10th century AD). Muqarnas are not carved as one piece. They are complex, fractal-like structures built by repeating, rotating, and stacking a few simple geometric "cells" (like squinches and triangles) according to a strict set of rules.

This historical practice is procedural generation. You design a simple set of rules and components, and an incredibly complex, beautiful, and stable architectural form emerges. A 'muqarnas generator' is a classic procedural modelling challenge.



Gaudí — Sagrada Família

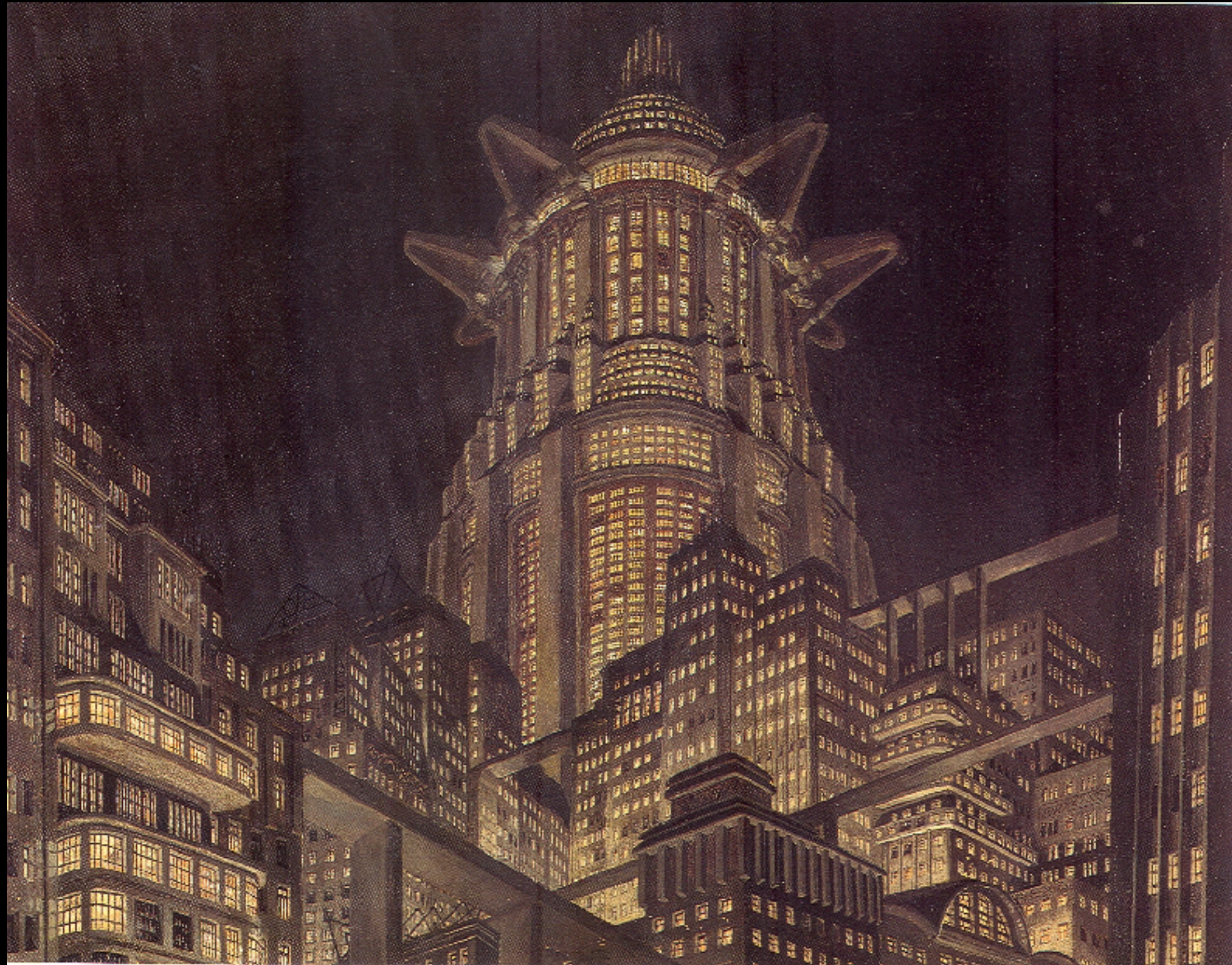


The Sagrada Família (1882—now) exemplifies the integration of nature and architecture, blurring the lines between the constructed and the organic.

Its intricate facades and towering spires draw inspiration from natural forms, fostering a dialogue between the urban and the natural environment. This masterpiece encourages to consider how architecture can reflect the spirit of a city and its relationship with the surrounding ecosystem.

[Virtual tour](#)

Fritz Lang — Metropolis

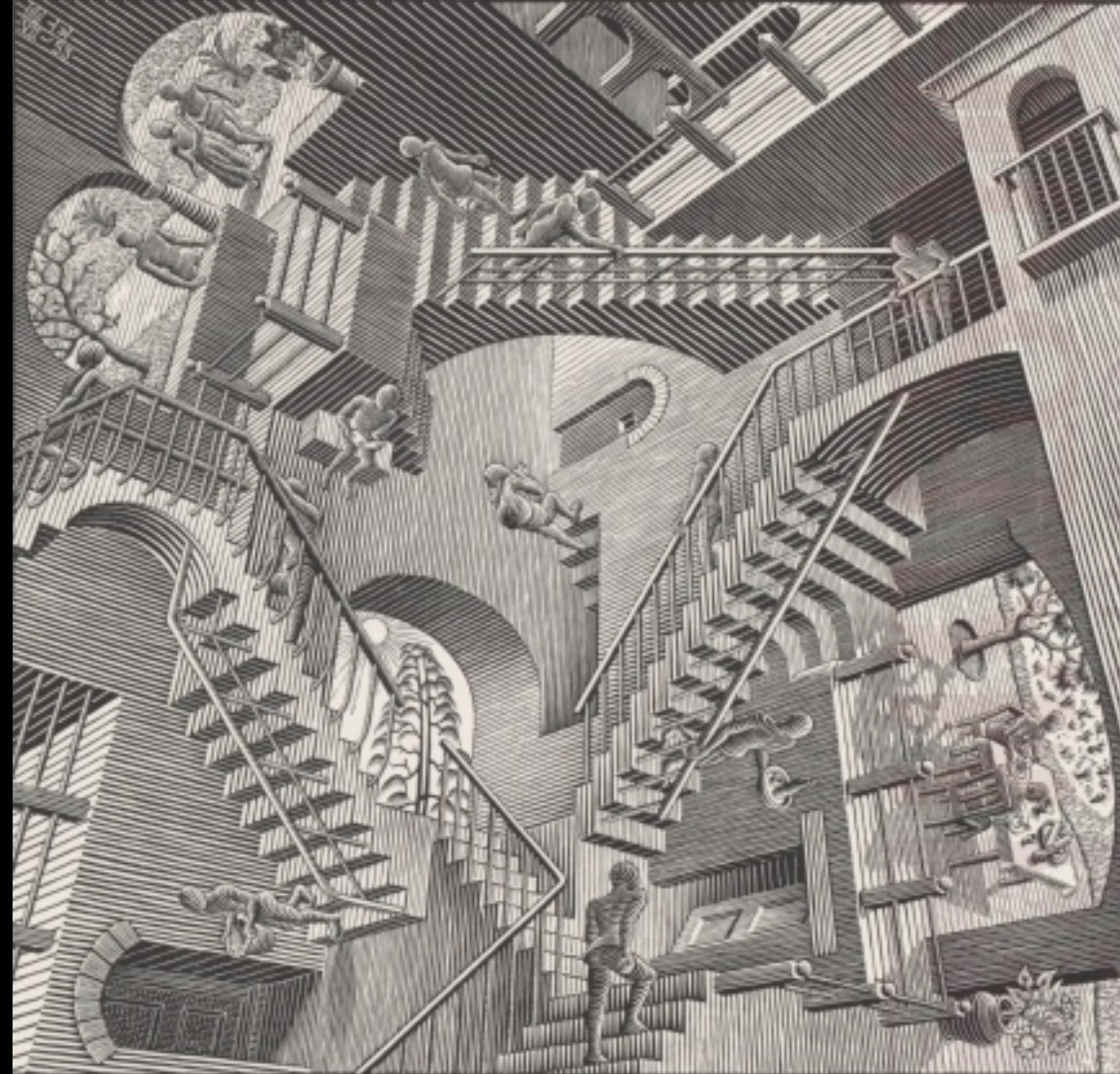
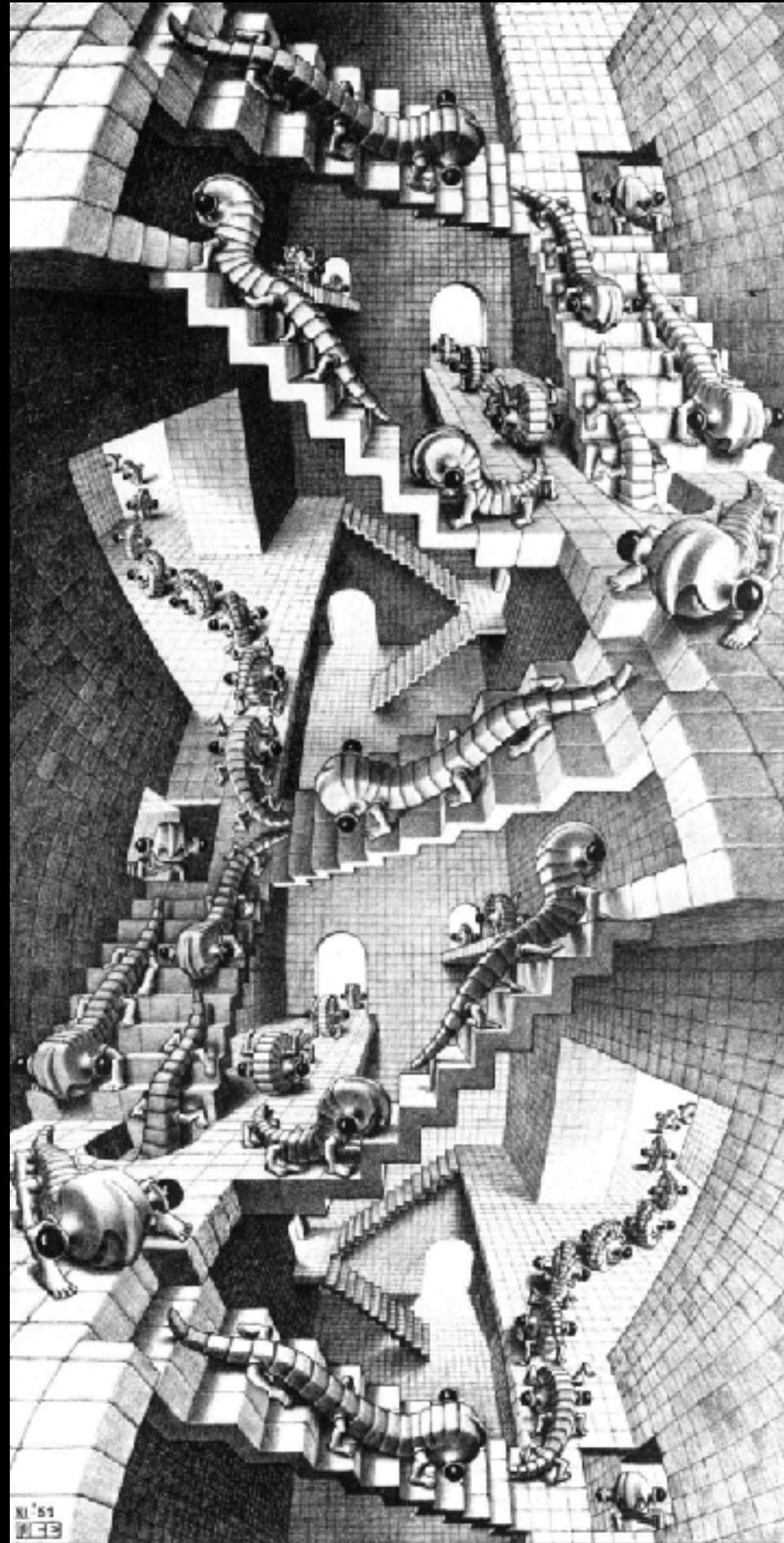


Silent science fiction film 'Metropolis' (1927) portrays a futuristic city sharply divided between the wealthy elite and the oppressed working class. The film's dystopian vision prompts critical reflections on the impact of social hierarchies and urban development on communities. It encourages to explore how visual narratives can convey complex socio-political messages about the dynamics of urban life.

Went on to inspire Blade Runner, 2001: A Space Odyssey and Star Wars.

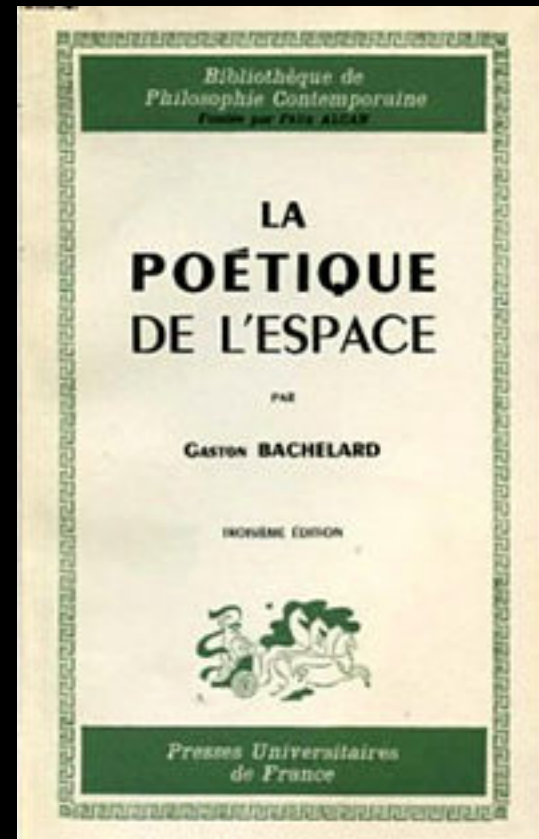
Cityscape scene

M.C. Escher



The work of Escher (1950s) has repeating patterns, clean lines, and impossible perspectives. Together they create images that look simple at first but become more complex the longer you look. Escher's designs turn flat surfaces into surprising spaces, where stairs loop back on themselves and shapes transform smoothly. Through this mix of clear geometry and visual tricks, he shows how an image can change the way we understand space and the structures around us.

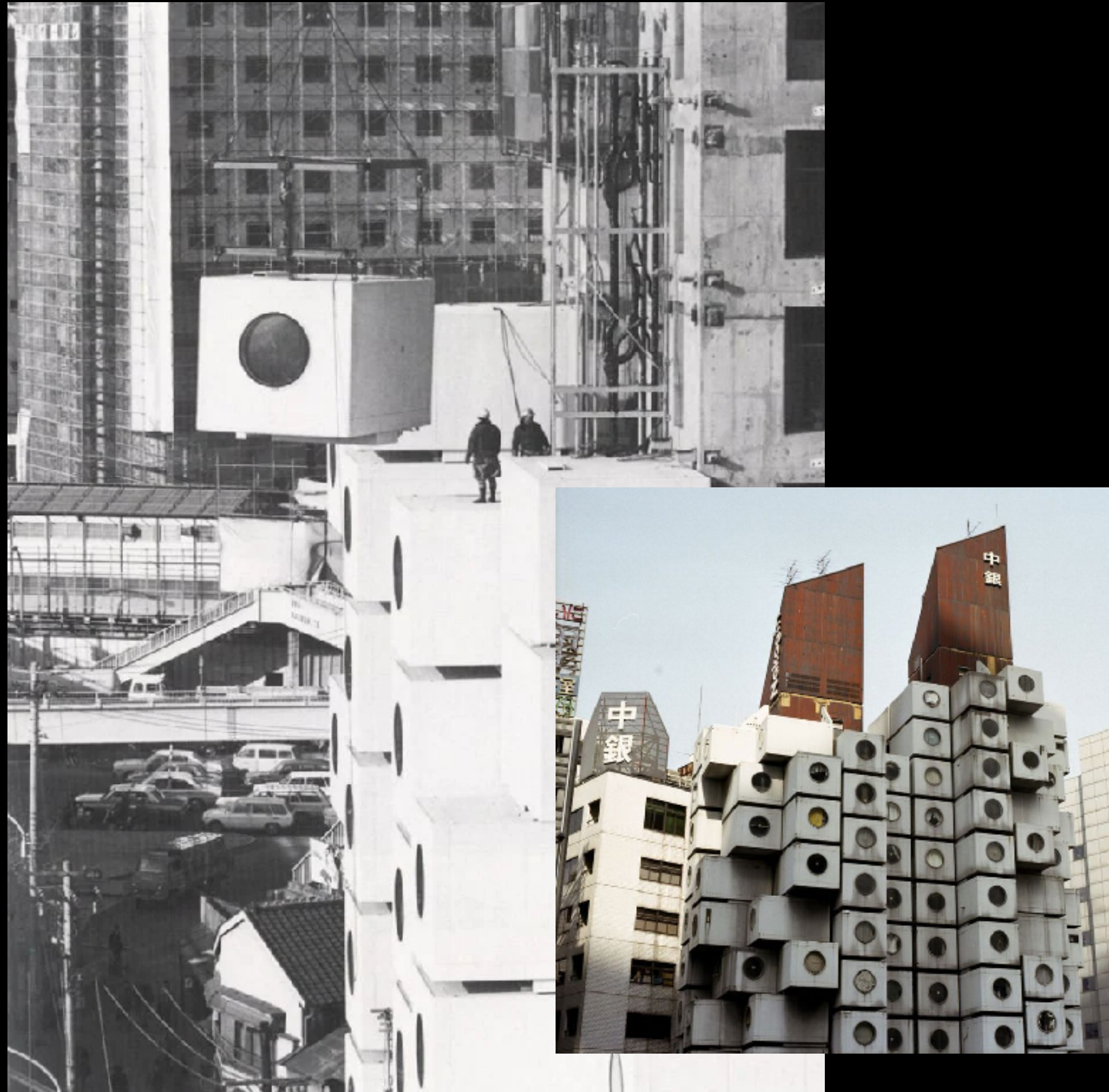
Gaston Bachelard — The Poetics of Space



In ‘La Poétique de l’Espace’ (1958) Bachelard applies the method of phenomenology to architecture, basing his analysis not on purported origins (as was the trend in Enlightenment thinking about architecture) but on lived experience in architectural places and their contexts in nature. He focuses especially on the personal, emotional response to buildings both in life and in literary works, both in prose and in poetry.

‘Sometimes the house of the future is better built, lighter and larger than all the houses of the past, so that the image of the dream house is opposed to that of the childhood home.... Maybe it is a good thing for us to keep a few dreams of a house that we shall live in later, always later, so much later, in fact, that we shall not have time to achieve it. For a house that was final, one that stood in symmetrical relation to the house we were born in, would lead to thoughts—serious, sad thoughts—and not to dreams. It is better to live in a state of impermanence than in one of finality.’

Metabolism



Metabolism (1959) was a post-war Japanese biomimetic architectural movement that fused ideas about architectural megastructures with those of organic biological growth.

During the preparation for the 1960 Tokyo World Design Conference, a group of young architects and designers prepared the publication of the Metabolism manifesto. They were influenced by a wide variety of sources, including Marxist theories and biological processes.

Their manifesto was a series of four essays entitled: Ocean City, Space City, Towards Group Form, and Material and Man, and it also included designs for vast cities that floated on the oceans and plug-in capsule towers that could incorporate organic growth.

Italo Calvino — Invisible Cities

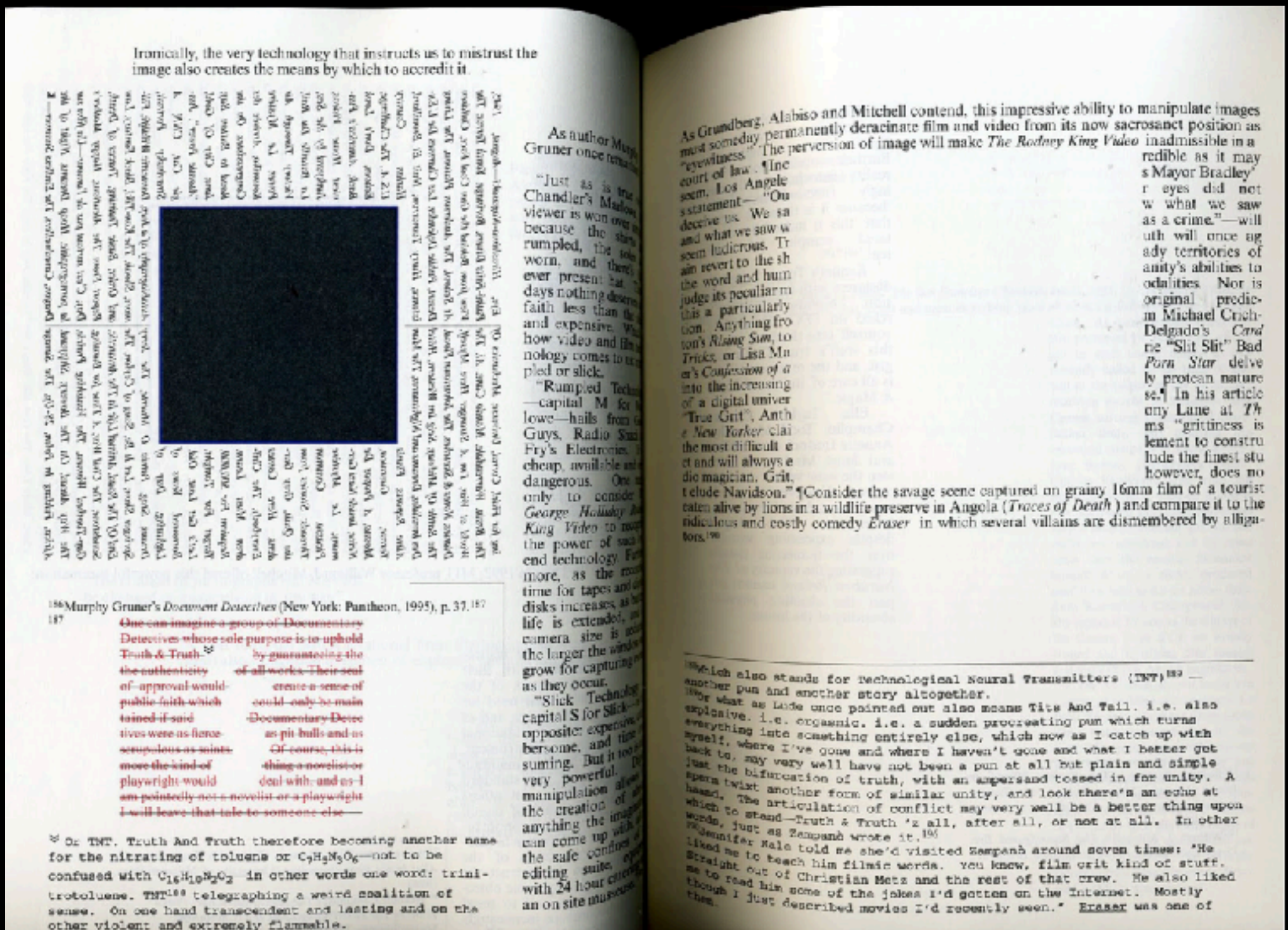


Italo Calvino's 'Invisible Cities' (1972) is a literary exploration of fantastical cities described by Marco Polo to Kublai Khan, reflecting on themes of memory, imagination, and the inherent subjectivity of urban experiences. Through Calvino's poetic prose, students can delve into the philosophical dimensions of urban spaces, contemplating how narratives can shape our understanding of cities as multifaceted, ever-evolving entities that exist beyond their physical structures.

Mark Z. Danielewski — House of Leaves

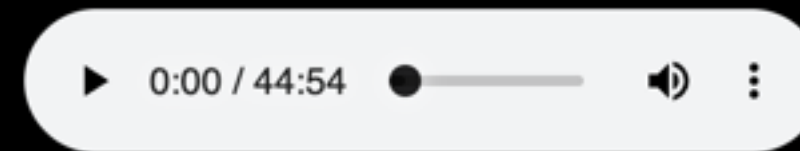
‘House of Leaves’ (2000) is a novel that defies conventional storytelling through its complex, labyrinthine narrative structure, exploring the uncanny tale of a family's encounter with an otherworldly, ever-expanding house. The intricate use of typography and formatting serves as a metaphor for the enigmatic and multifaceted nature of urban spaces, inviting readers to question the boundaries between reality and fiction, just as cities can blur the lines between physicality and imagination.

This work prompts to contemplate how narrative techniques can mirror the elusive and mutable essence of cities, emphasizing the role of storytelling in unraveling the hidden layers of urban environments.



Vito Acconci — The Bristol Project

‘The Bristol Project’ (2001) is a sonic walkthrough of an imagined, futuristic city. In this audio-only piece, Vito Acconci uses his voice to guide the listener, describing fantastical buildings and spaces (including some from his own architectural studio) as if they were real. The listener is meant to experience this imagined architecture superimposed over the real city. Acconci ‘builds’ an entire city using only language. It's a ‘space-producing narrative’.



Alicja Kwade

Alicja Kwade's work (2019) explores how objects, materials, and spaces can change meaning when we look at them from different angles. She often uses mirrors, stones, metal frames, and simple geometric forms to create installations that feel both scientific and poetic. These setups play with balance, reflection, and repetition, making the viewer question what is real and what is only an illusion. Through these clear and carefully arranged structures, Kwade shows how our understanding of space and reality depends on perspective, inviting us to see familiar materials in a new way.



Monument Valley

— ustwo games

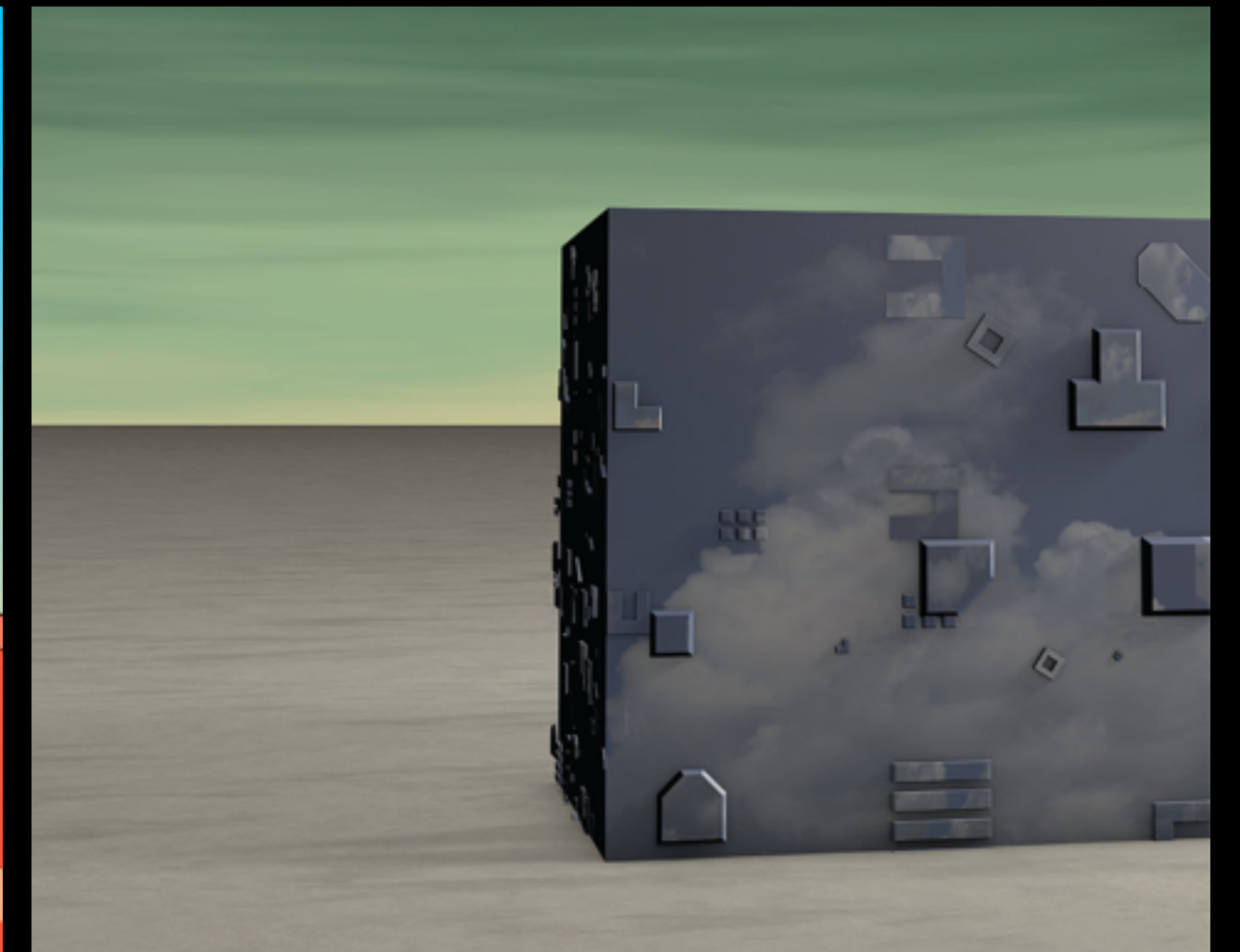
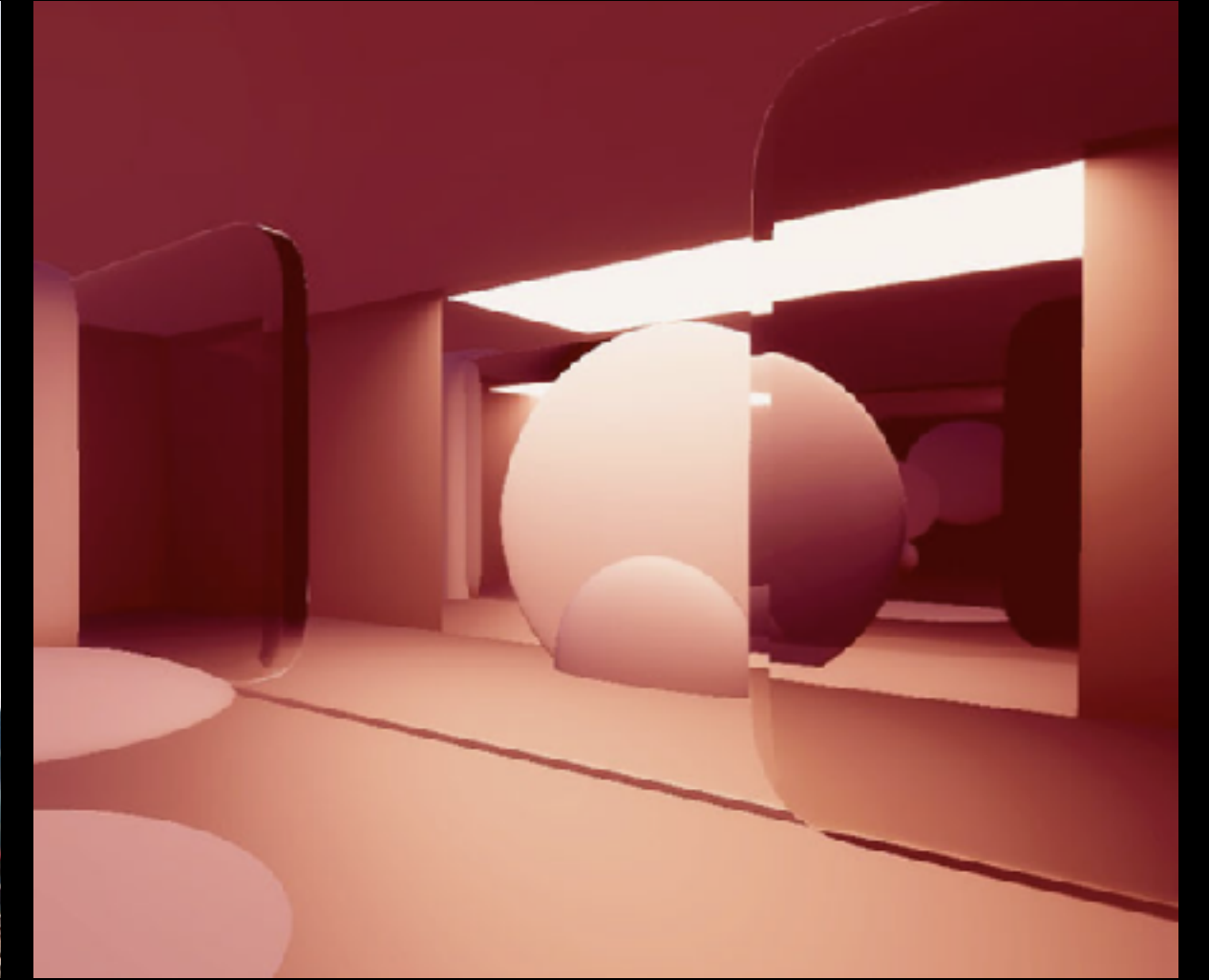
Monument Valley (2014) by ustwo games feels like a wandering through a calm dream. The player explores impossible architecture rendered through illusionary geometry. These quiet, Escher-like worlds show how perception itself can shape a journey, turning simple movement into a small act of discovery.

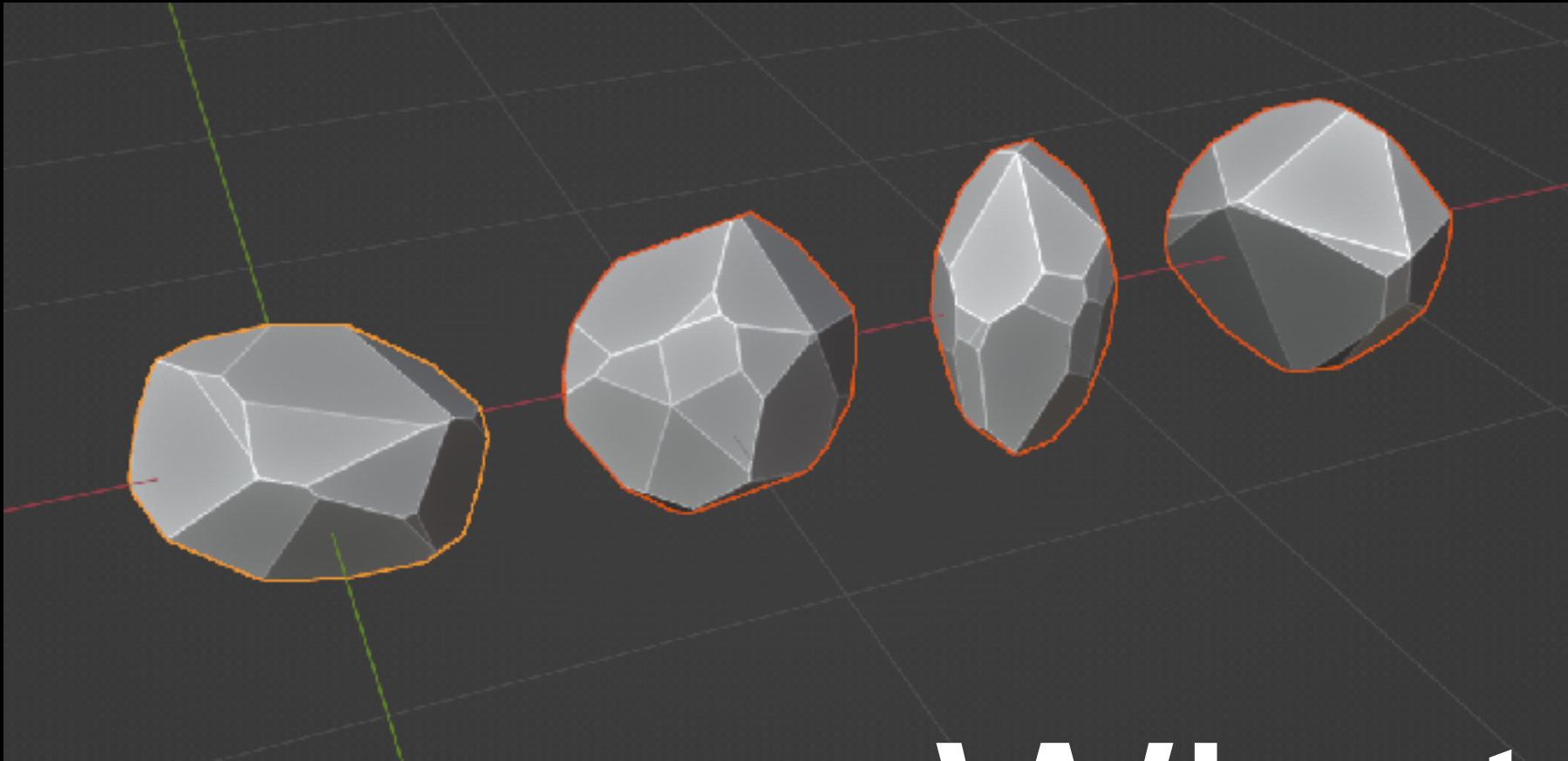
Trailer

Available on iOS / Android / Windows / ... for less than a flat white



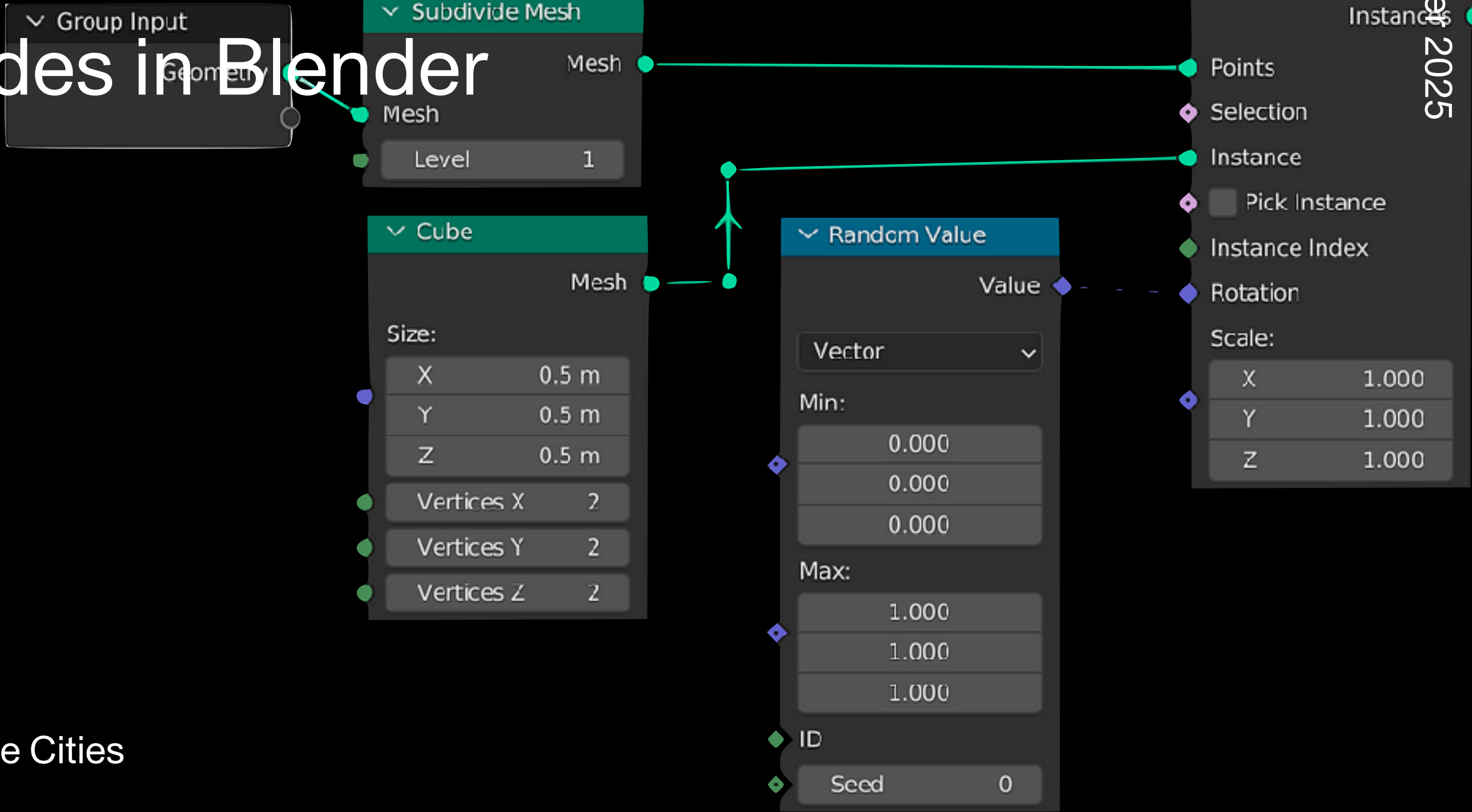
Are.na board — Invisible Cities

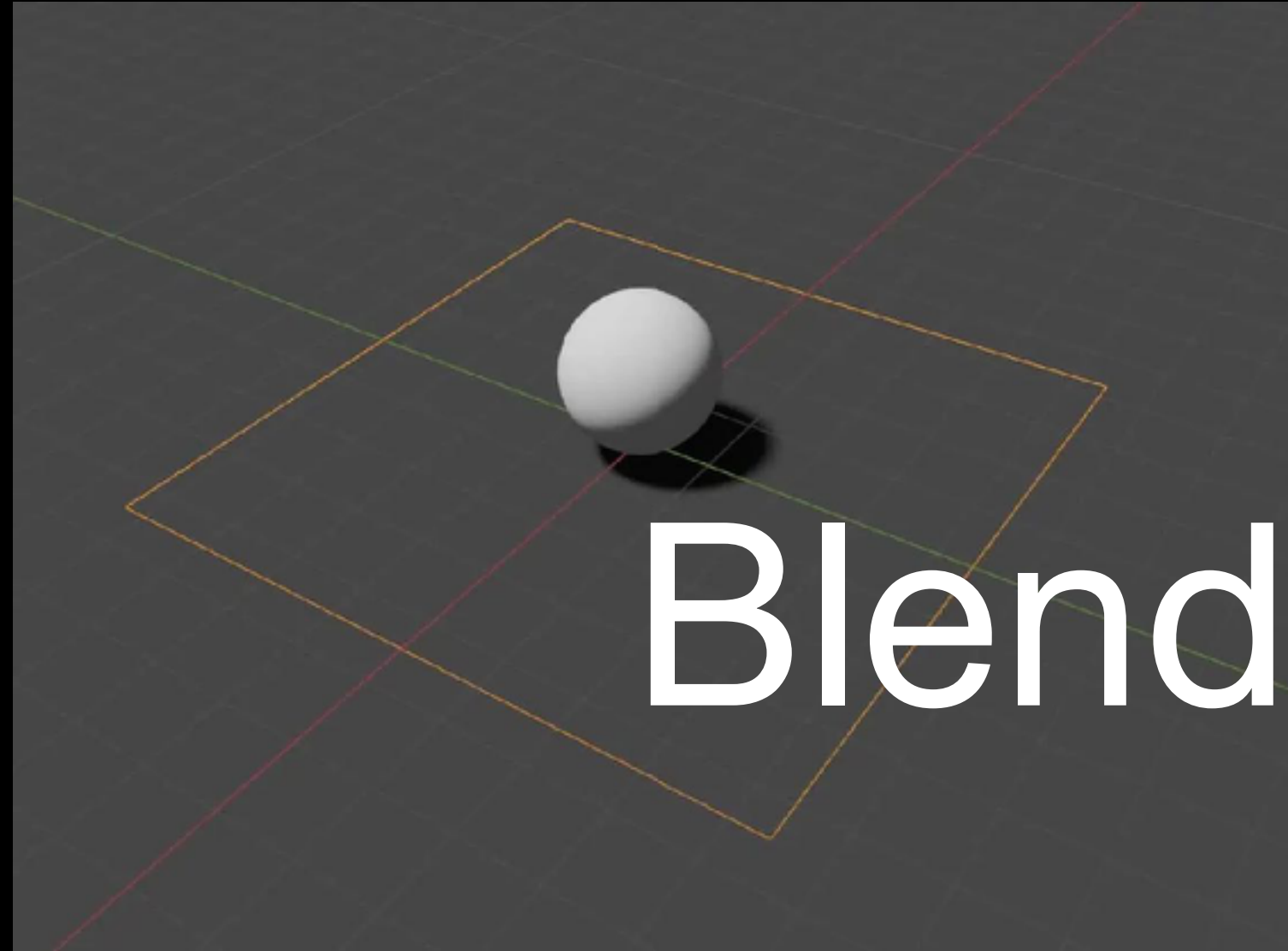




What you will learn

Geometry Nodes in Blender





Blender: the basics

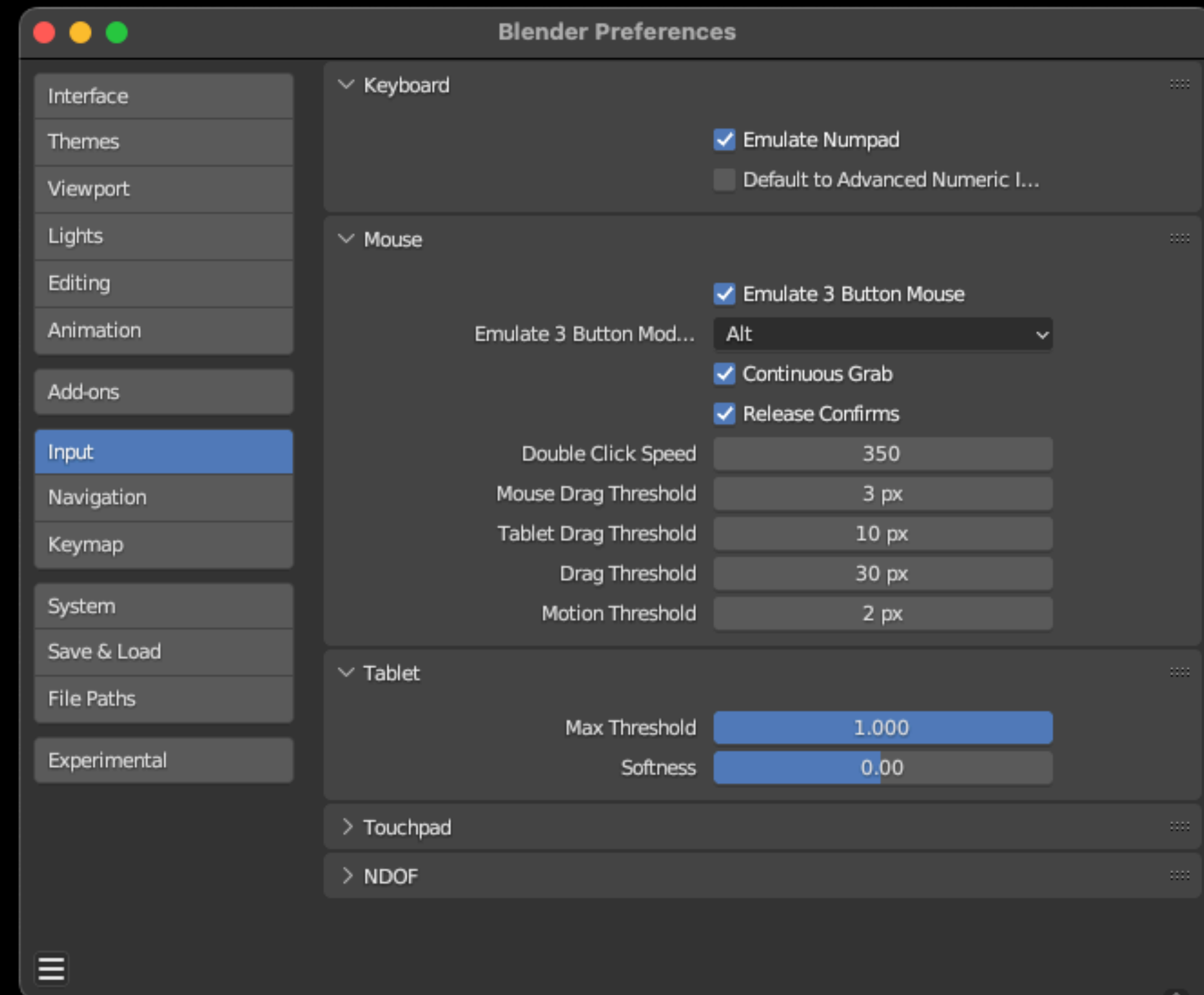
The interface

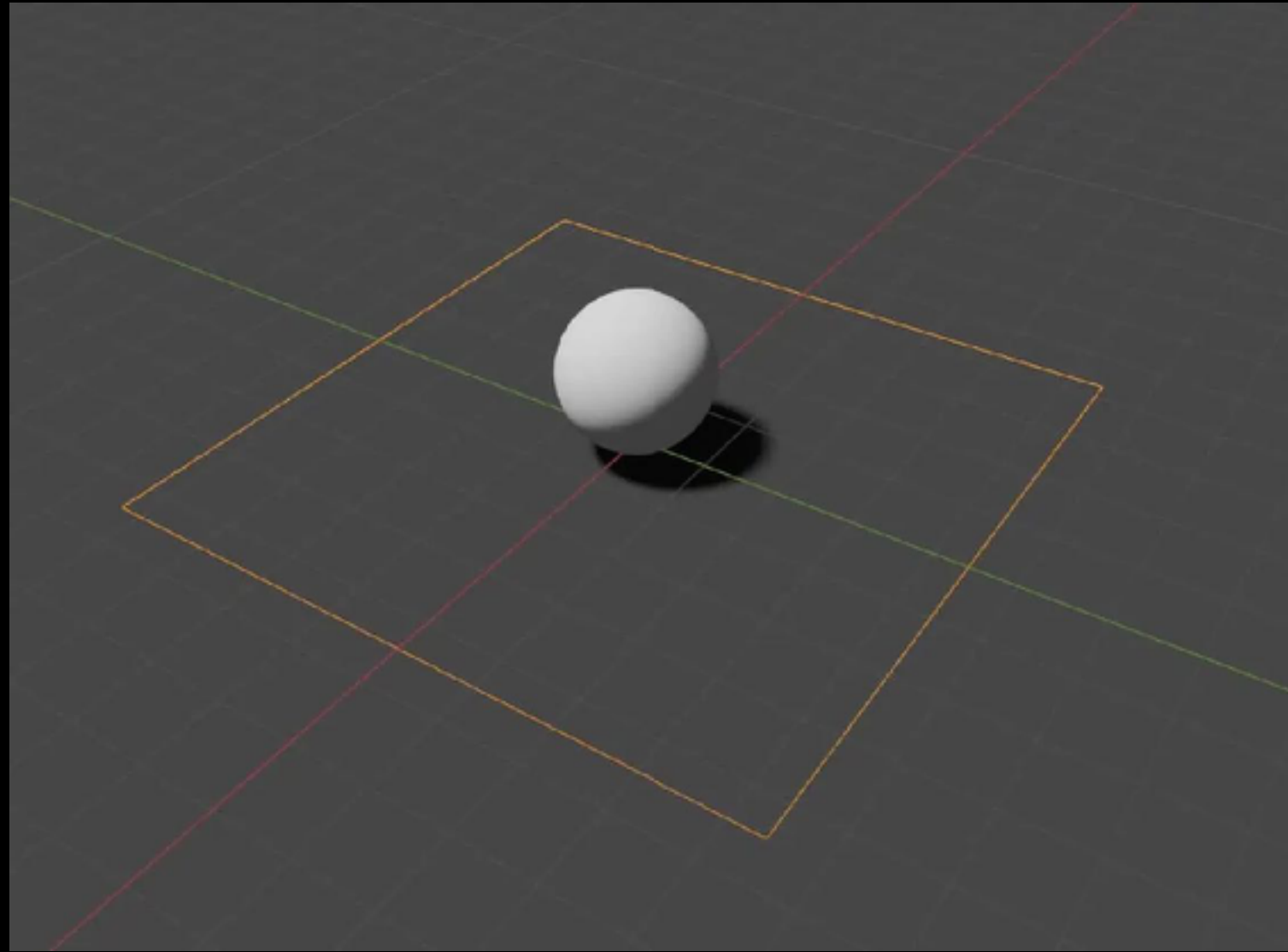
The workflow

Basic modelling

Check Blender Preferences!

1. Input: Emulate **3 button mouse**
2. Input: Emulate **Numpad**
3. Navigation: **Orbit & Pan Auto Depth**
4. System: use **GPU**





The Workspace
Viewport Control
Grab, Rotate, Scale
Axis Snapping
Object Mode & Edit Mode
Vertices, Edges, Faces
Render Modes

Abstract Sculpture

Starting from a cube

Using Edit Mode

Scale

Move

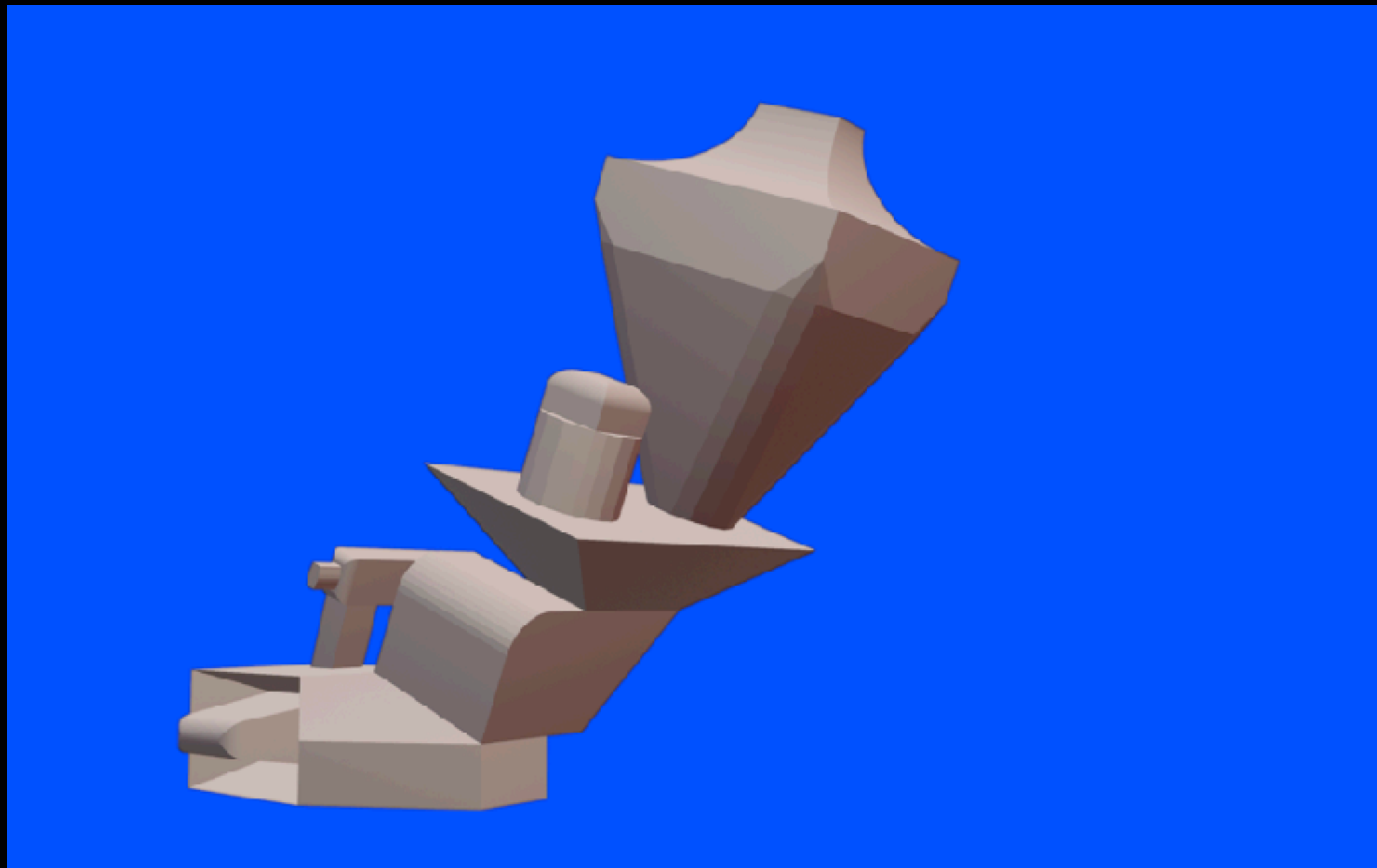
Extrude

Snap

Inset

Rotate

Bevel



What are Geometry Nodes?

(And why would you use them)

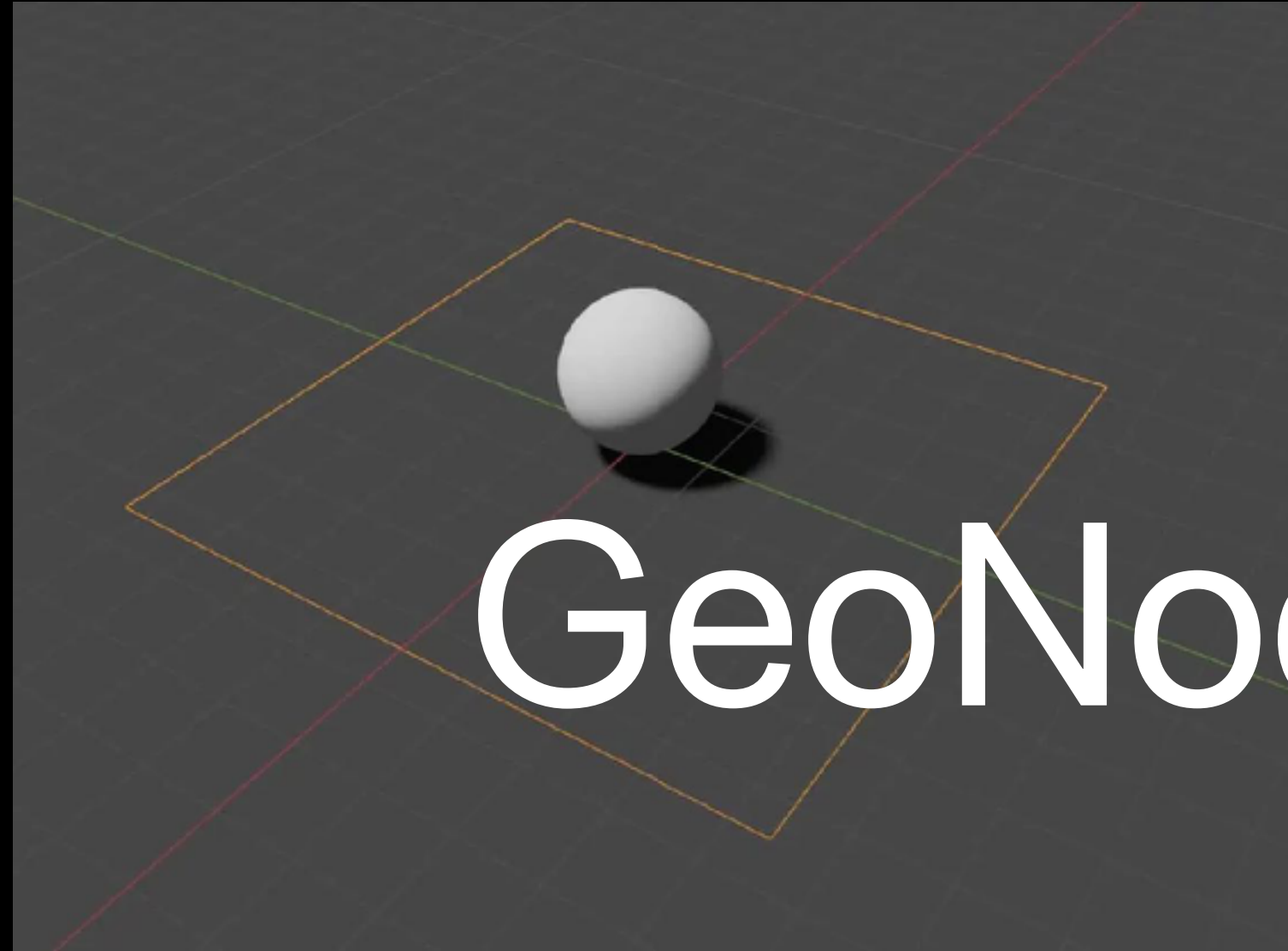
Geometry nodes are a basic form of visual programming in Blender
They allow you to do generative design

Now, let's watch a short video about geometry nodes in Blender



In this workshop, you will learn how to use geometry nodes to build procedural designs in Blender.

At the end, you will bring your design into the physical world.



GeoNodes: the basics

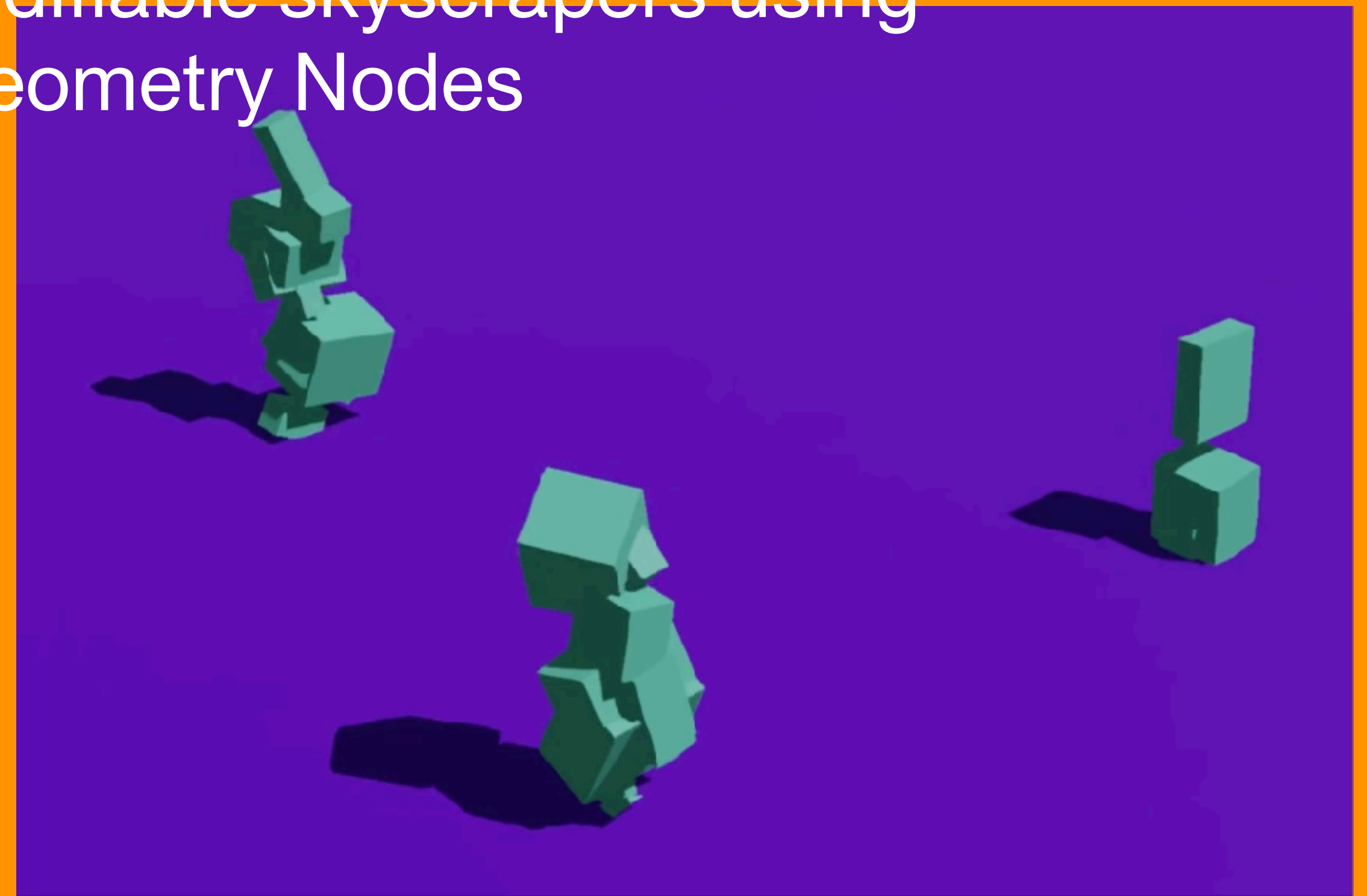
Essential nodes

Starting our first node tree

Introducing material

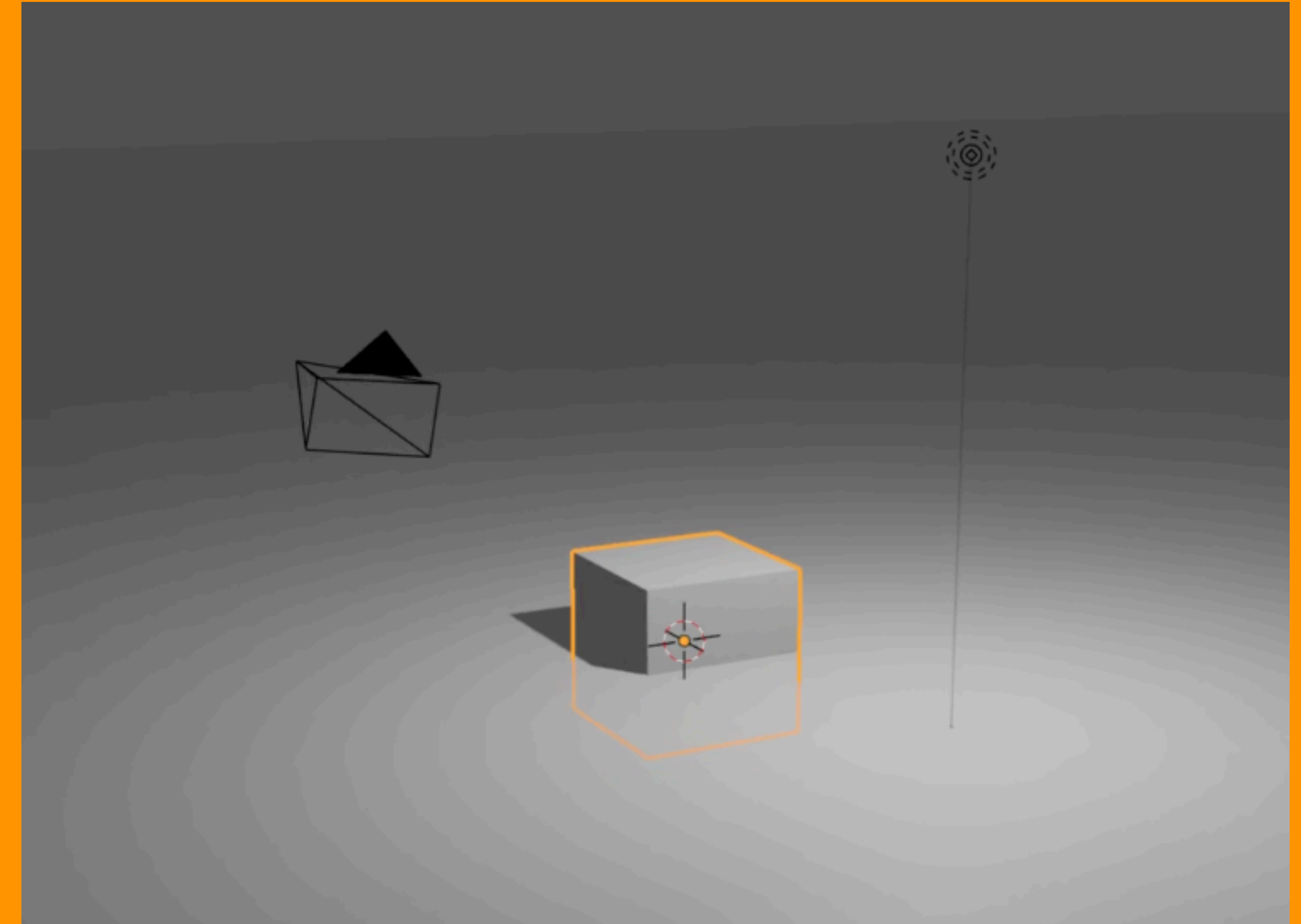
Tutorial: Quick Skyscrapers

Building generative, modifiable skyscrapers using
Blender's Geometry Nodes



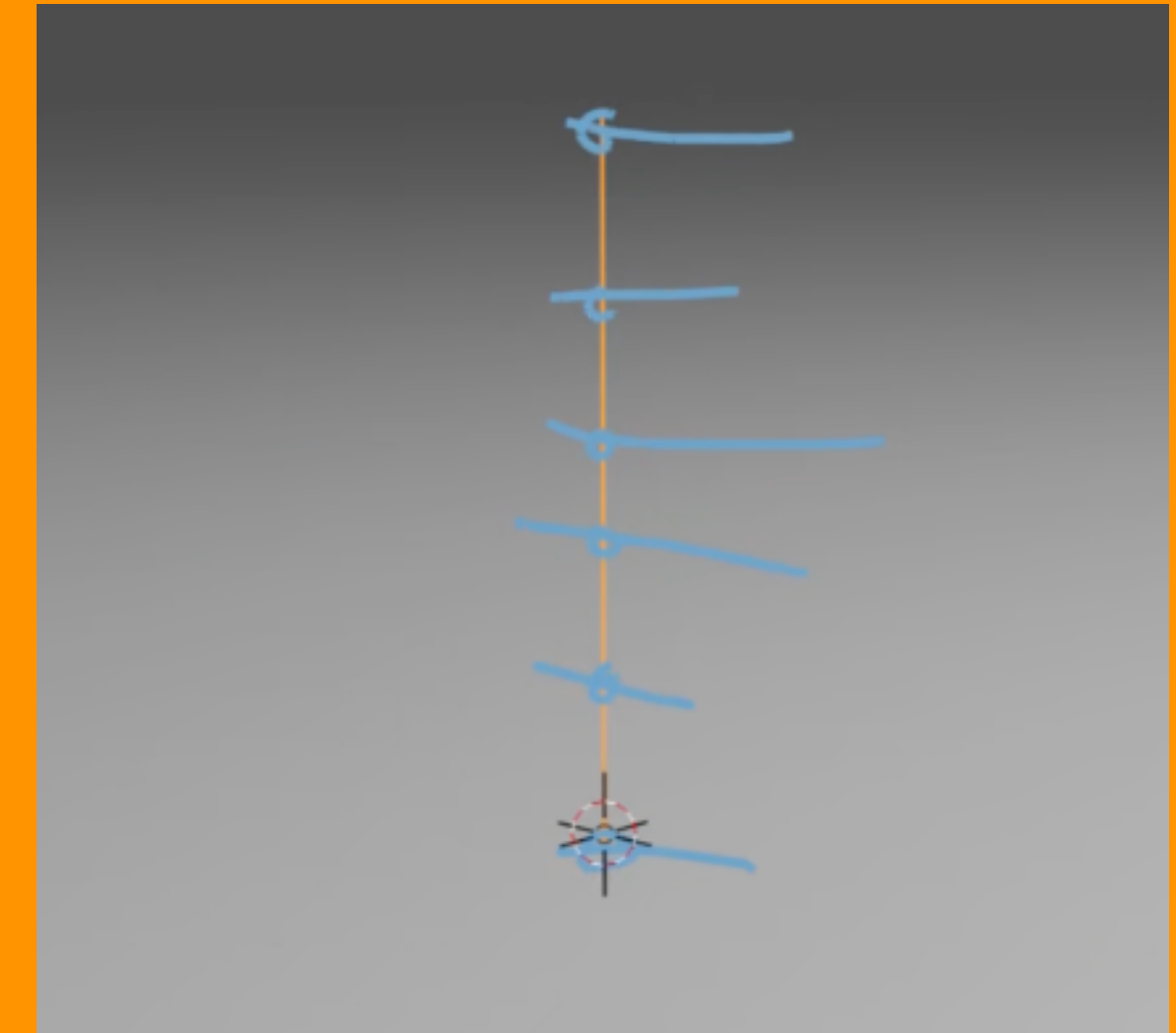
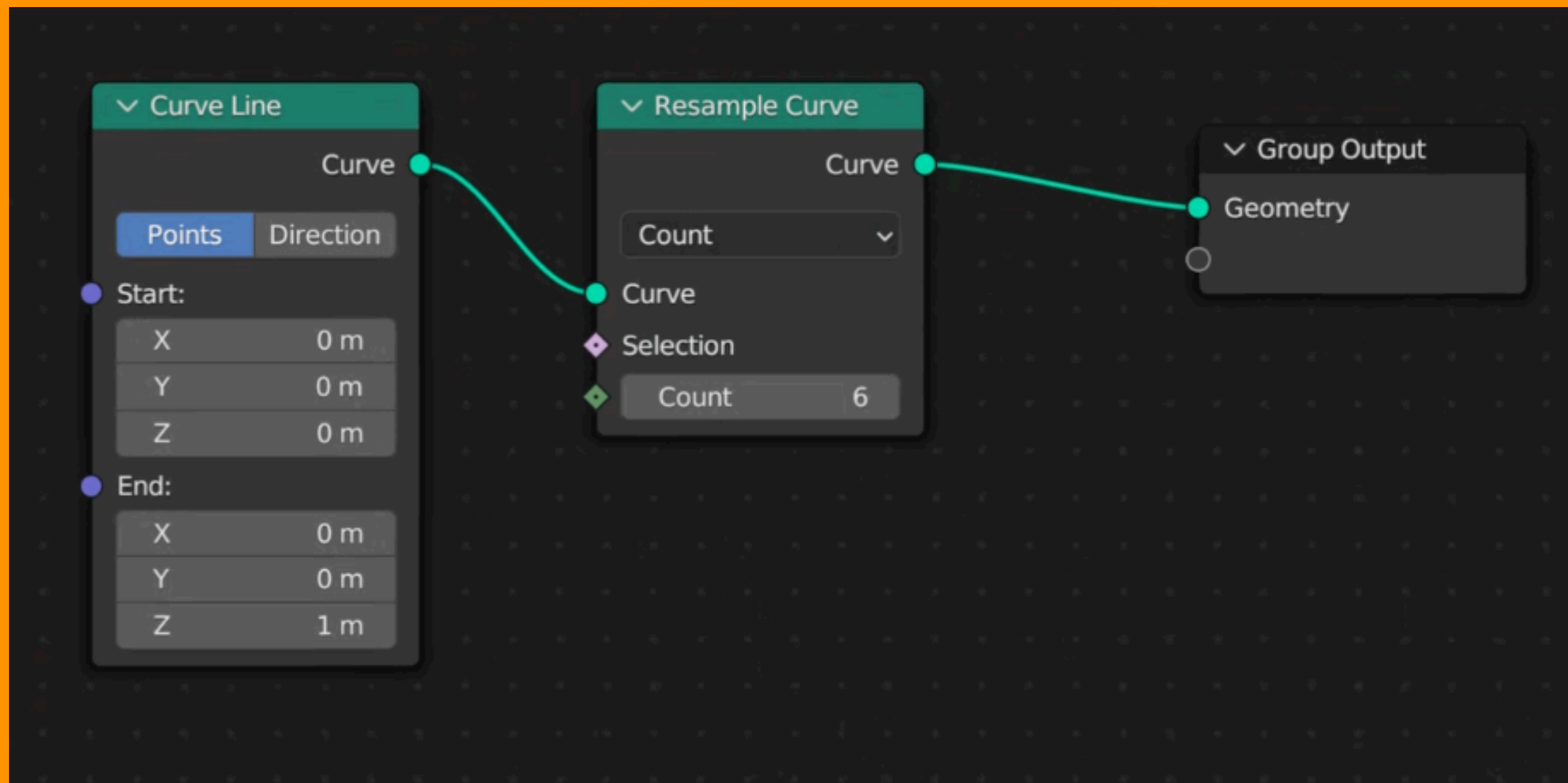
Setup

1. Create a **new file**
2. Add > Mesh > **Plane**
3. **Scale** plane (100x100)
4. **Rename** Cube to Skyscraper
5. **Rename** Plane to Ground Plane



Adding Geometry Nodes and a curve

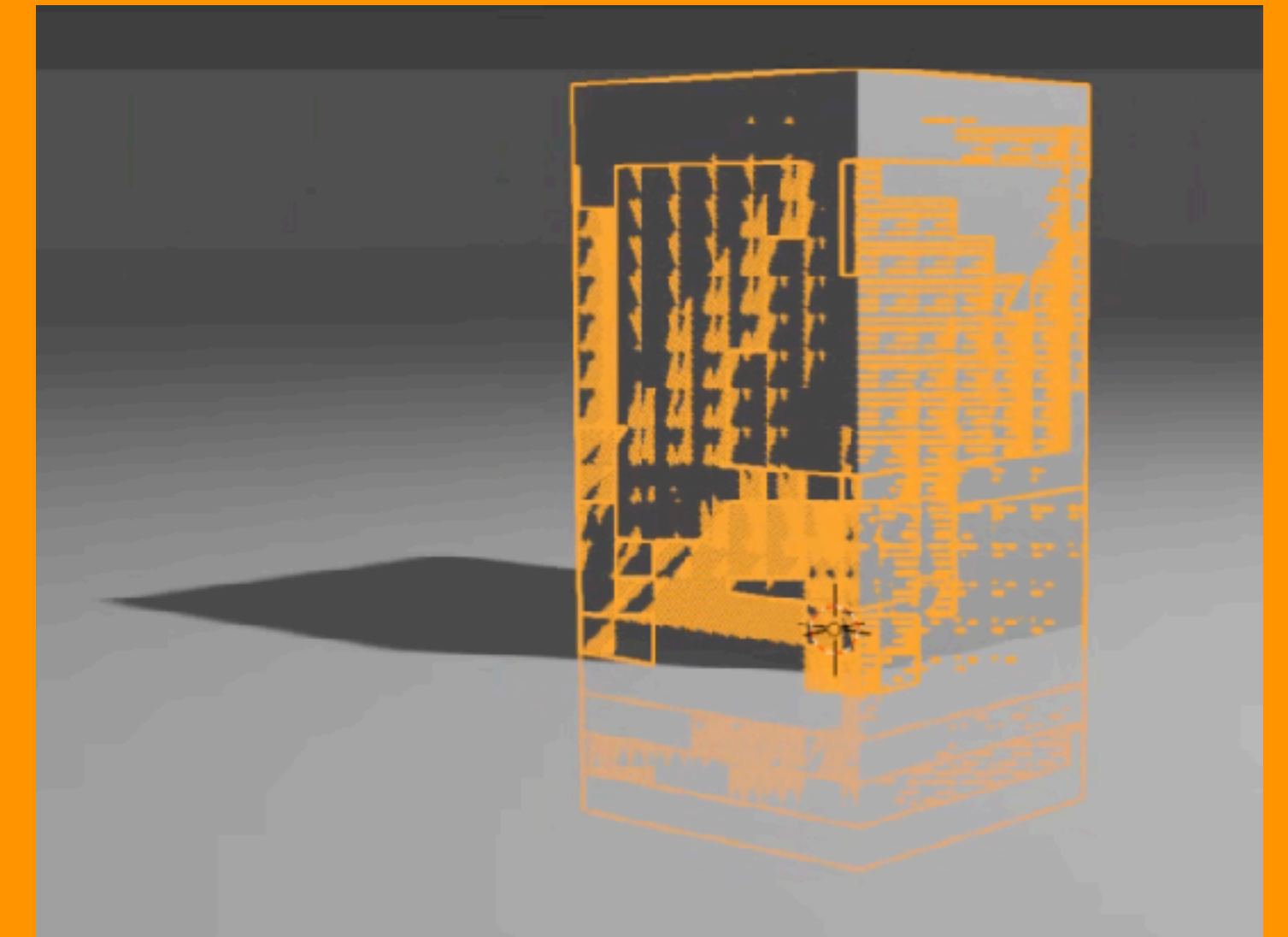
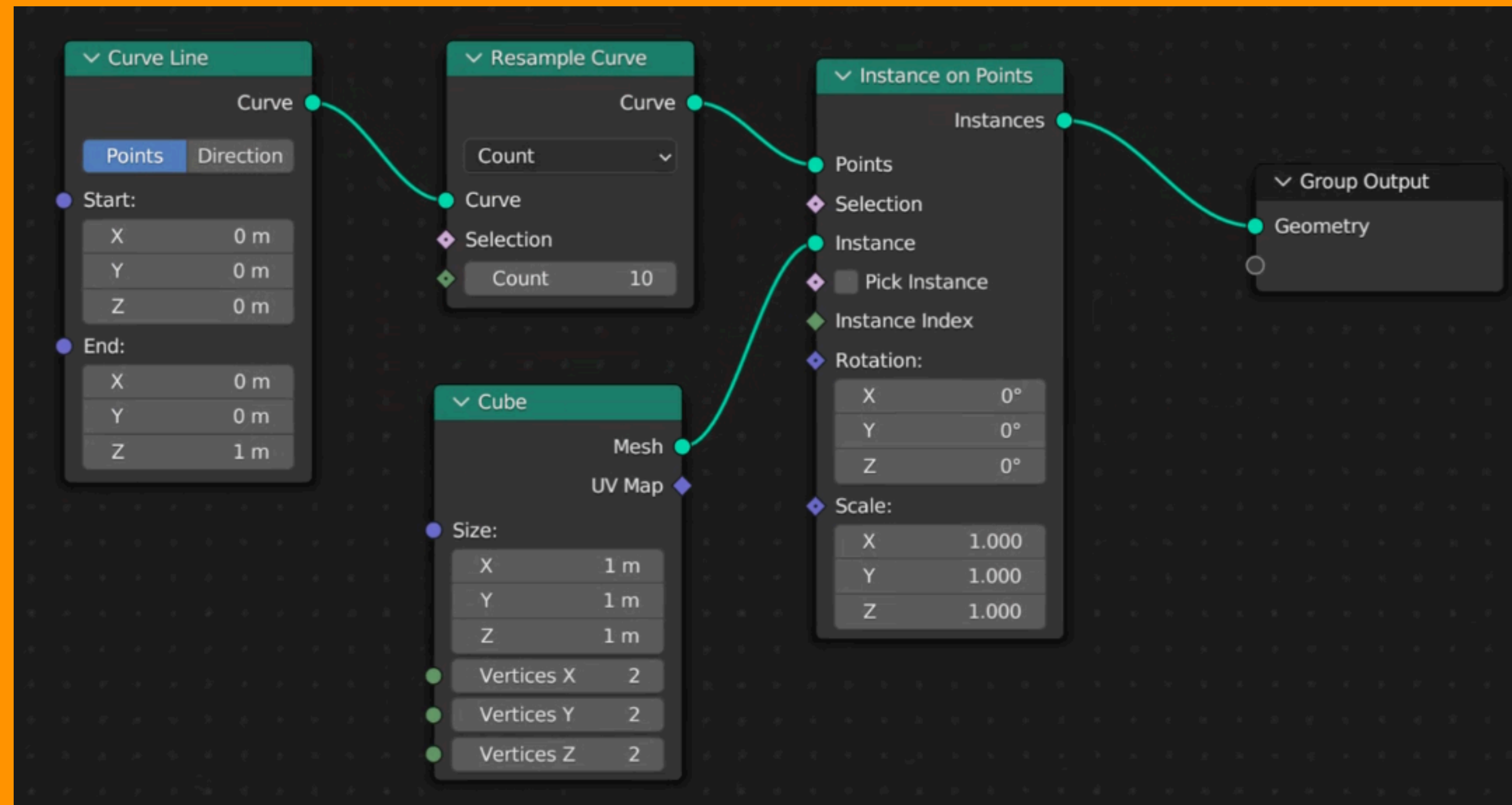
6. Change **Timeline** panel to **Geometry Nodes**
7. Add new geometry node
8. Delete **Group Input** node (x key)
9. Add **Curve Line**
10. Add **Resample Curve**



Adding a cube to each segment

10. Add Cube node

11. Add Instance on Points

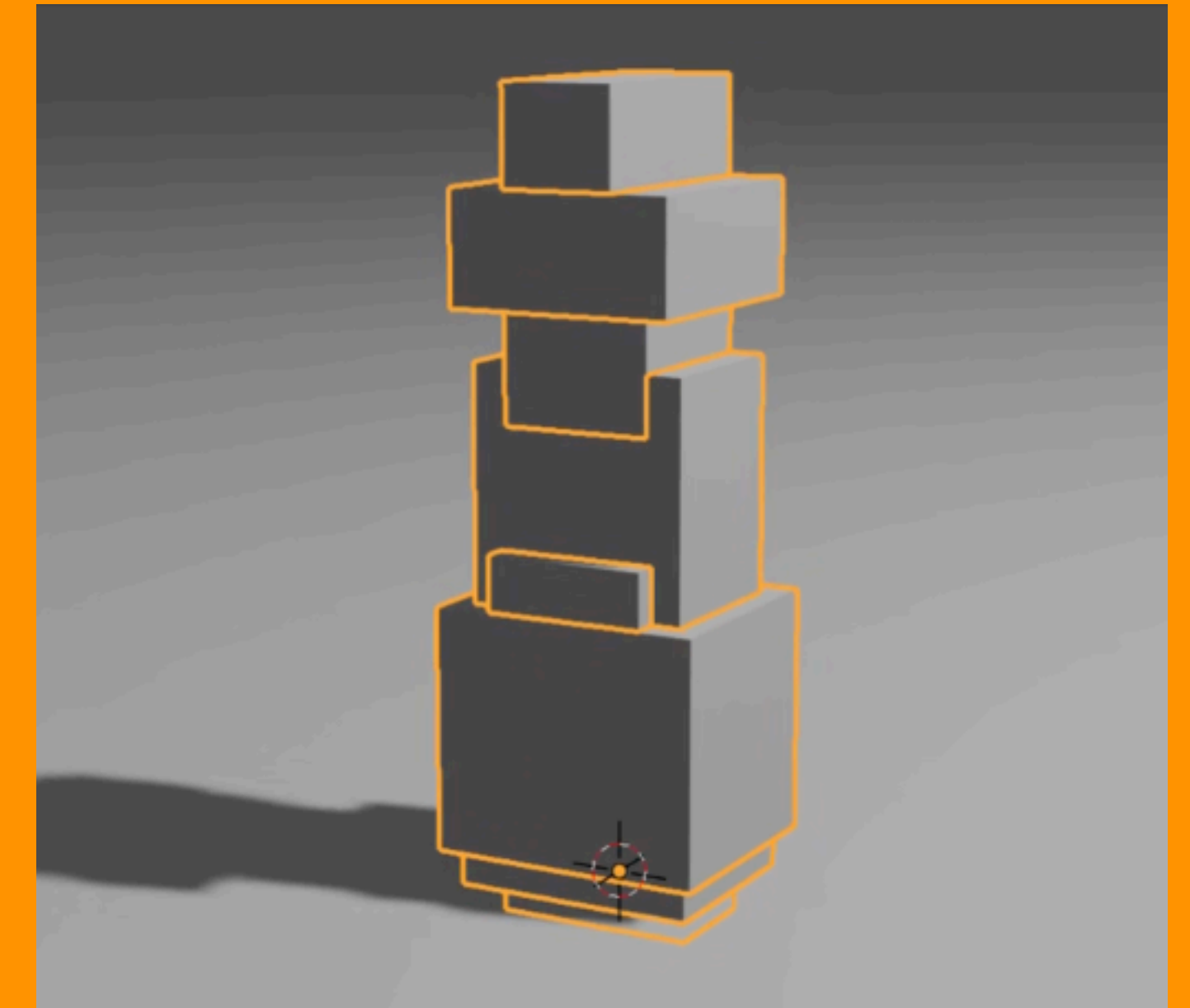
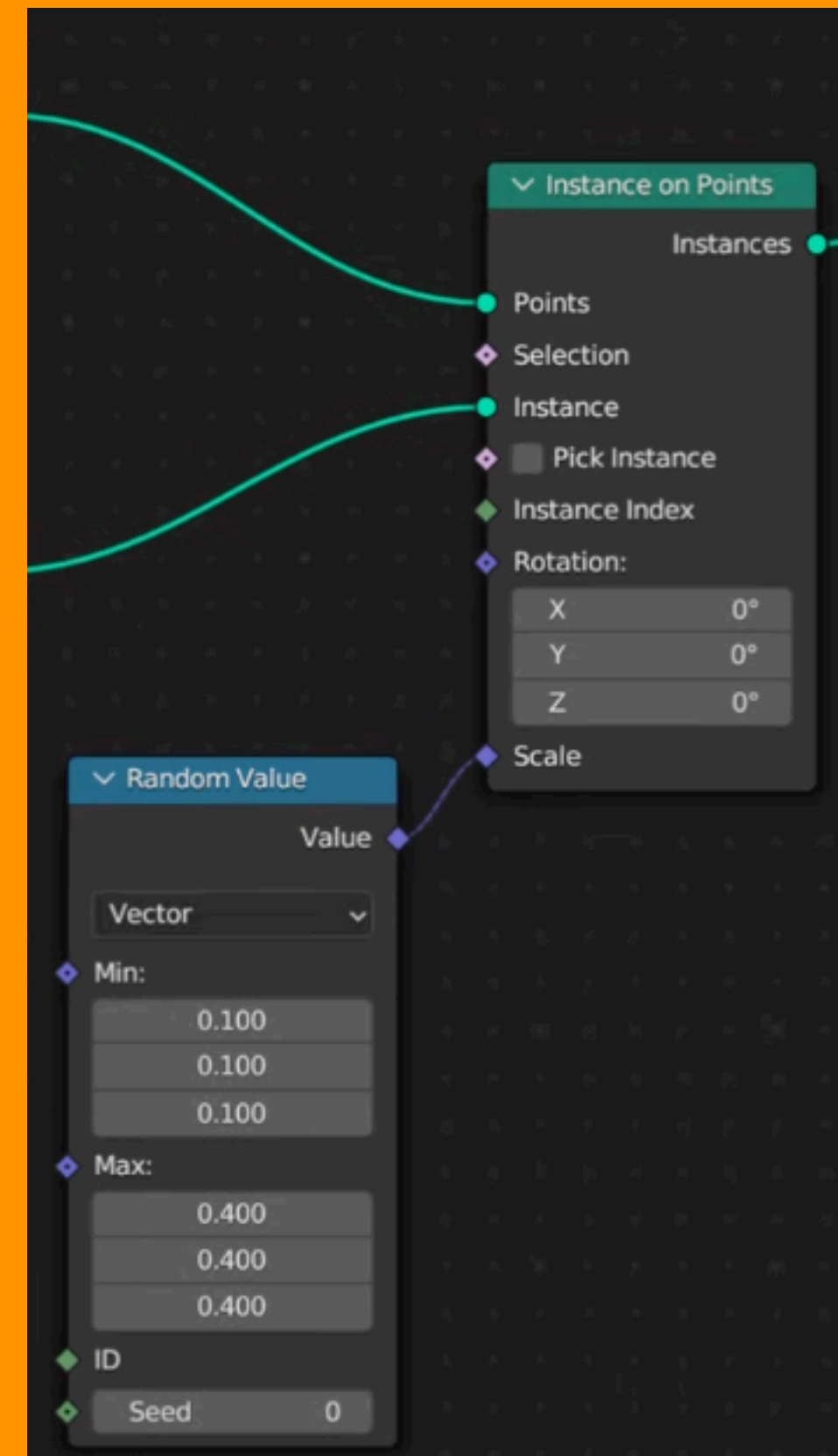
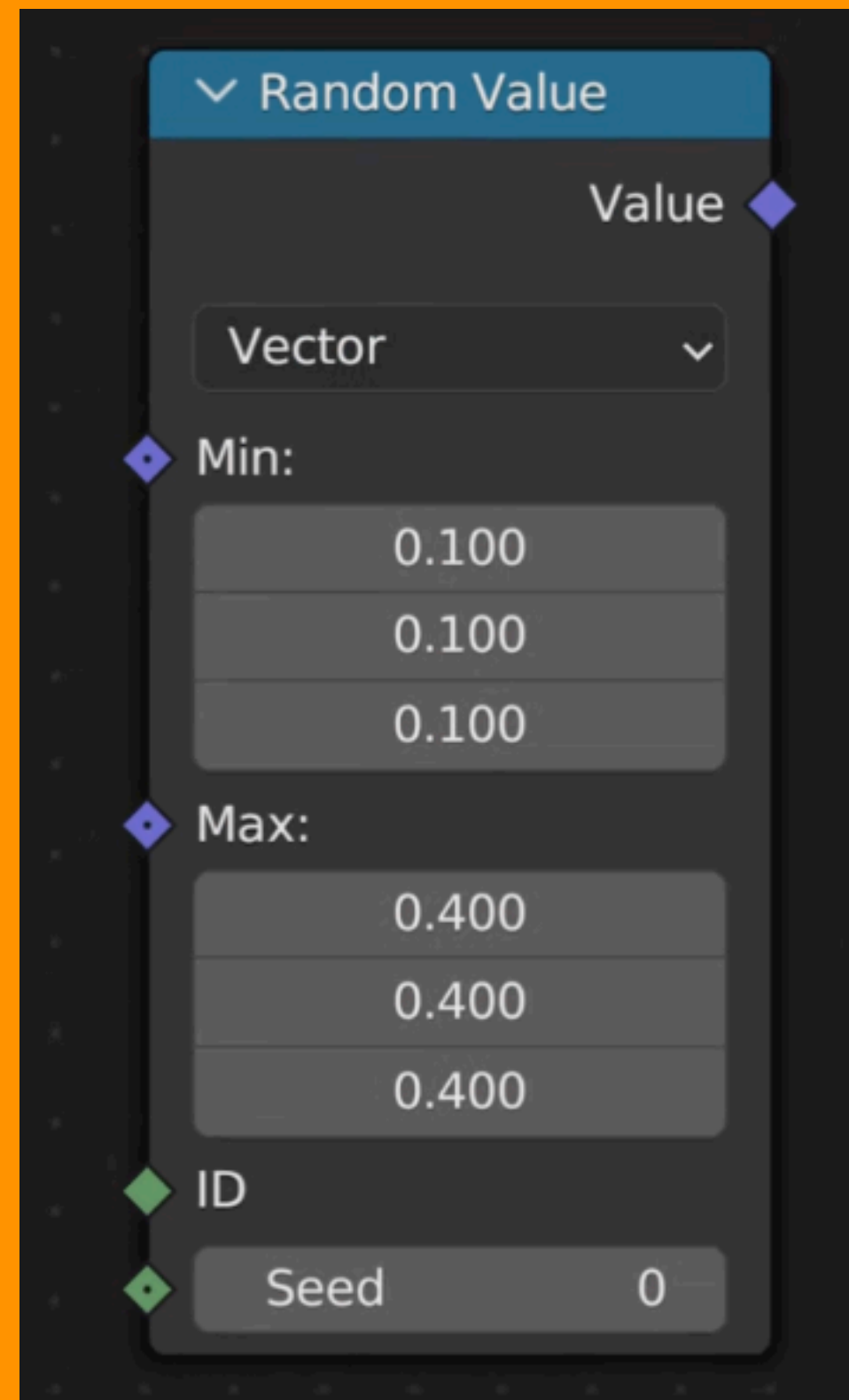
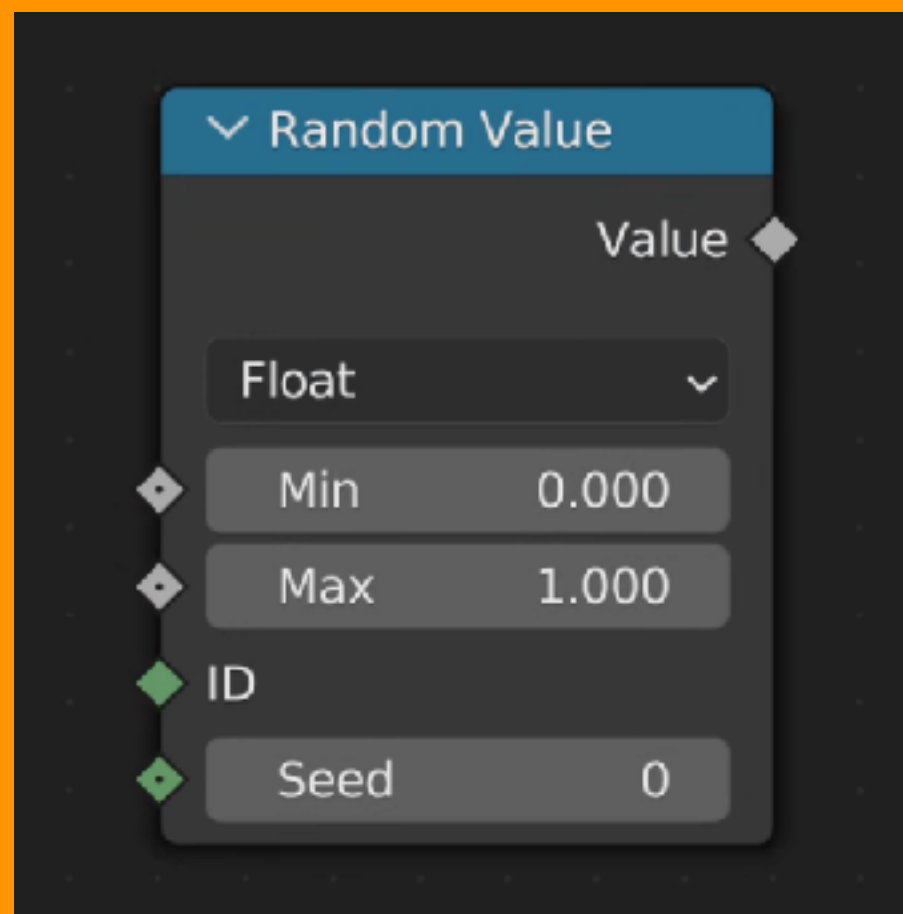


Adding randomness

12. Add **Random Value**; set it to **Vector**

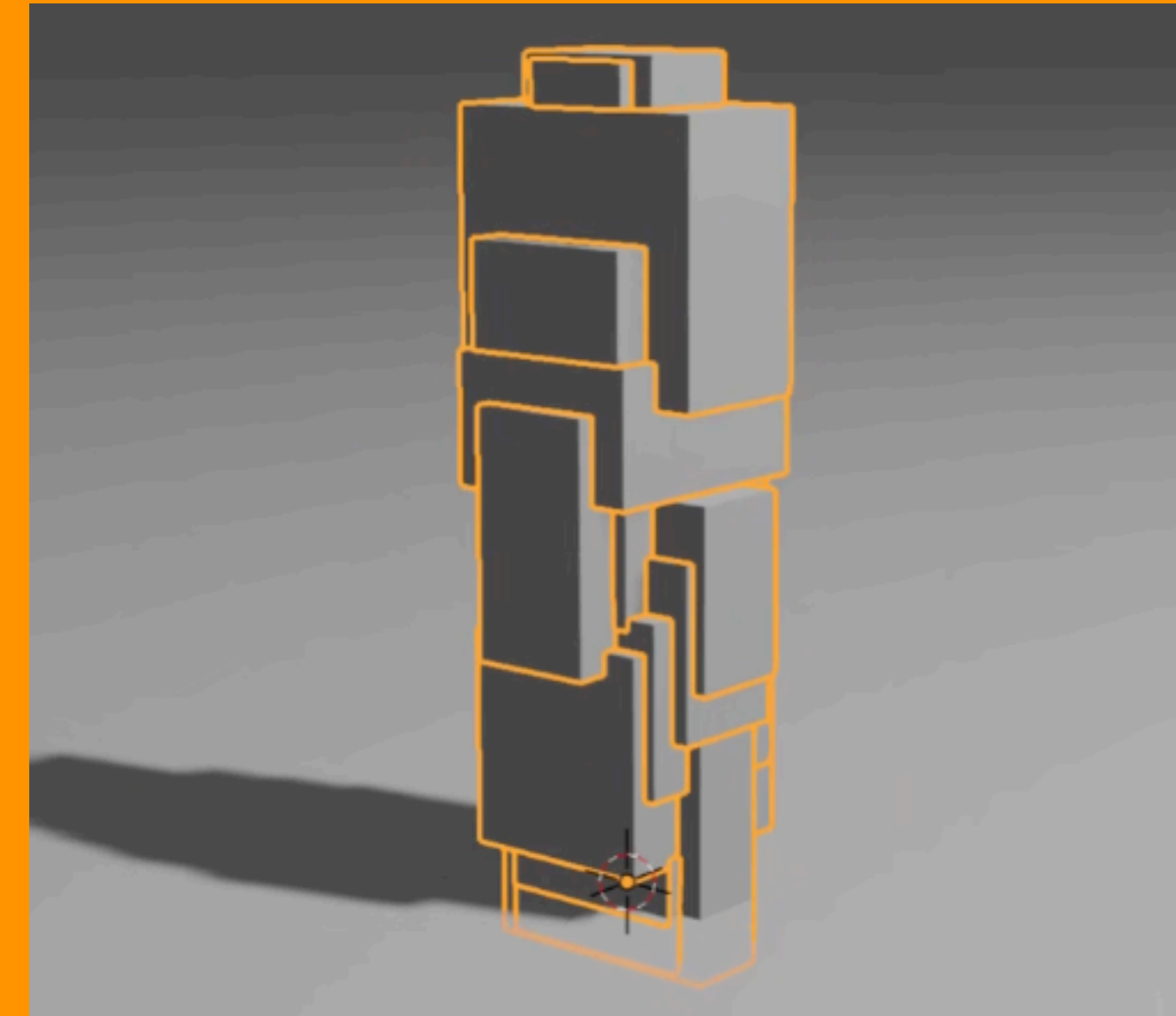
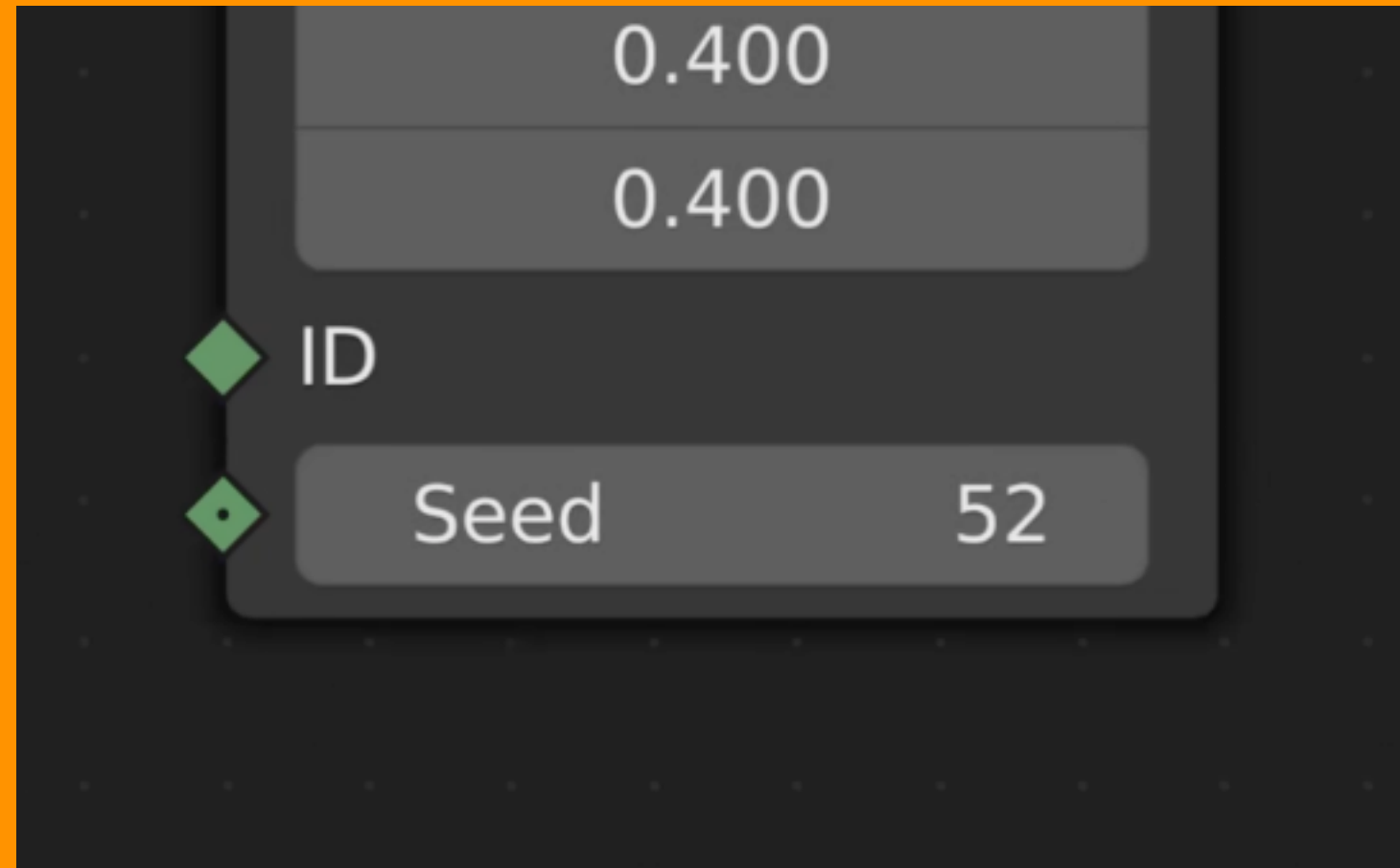
13. Add **Instance on Points**

14. Connect **Random Value** to **Scale**



Changing the seed

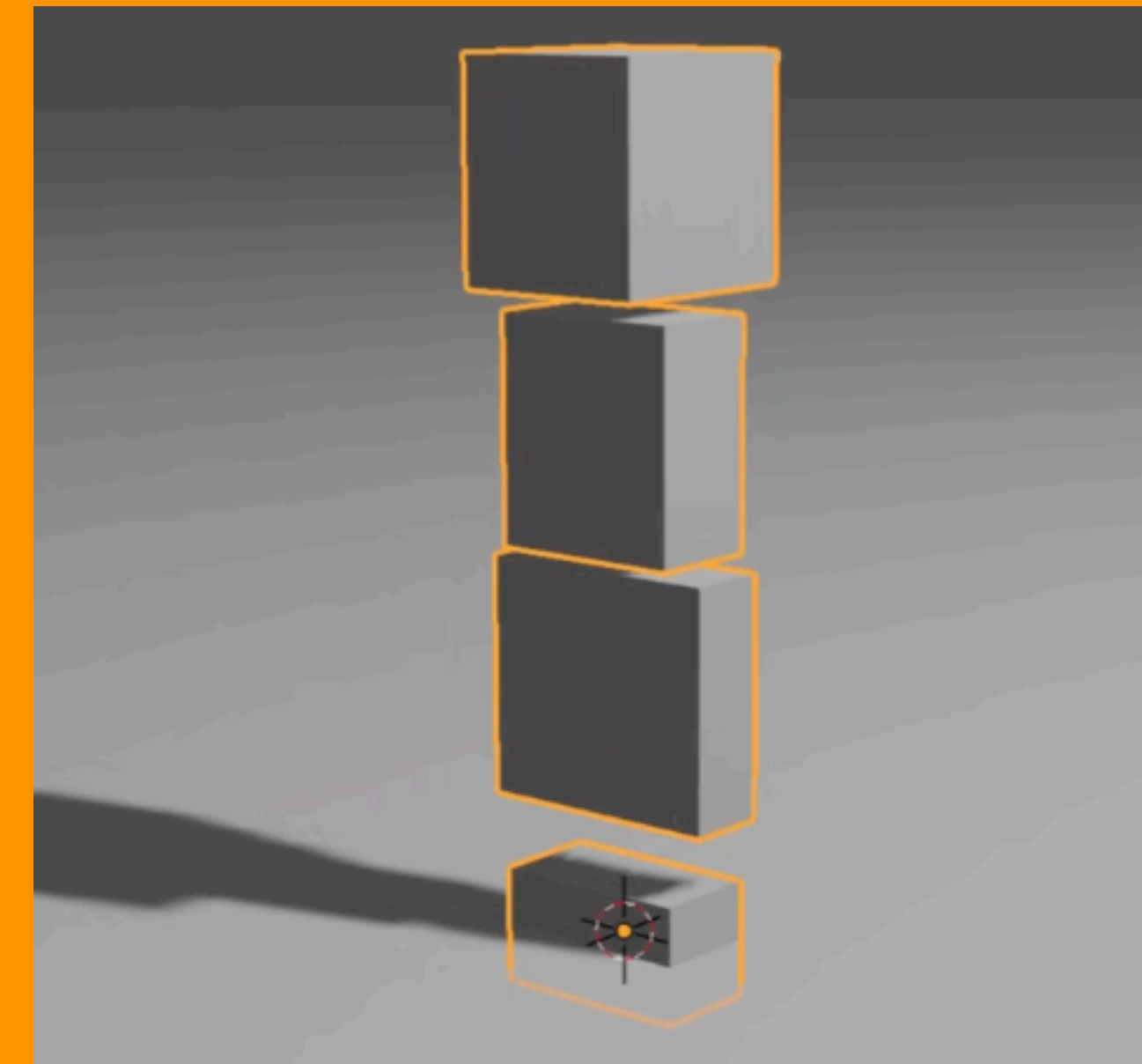
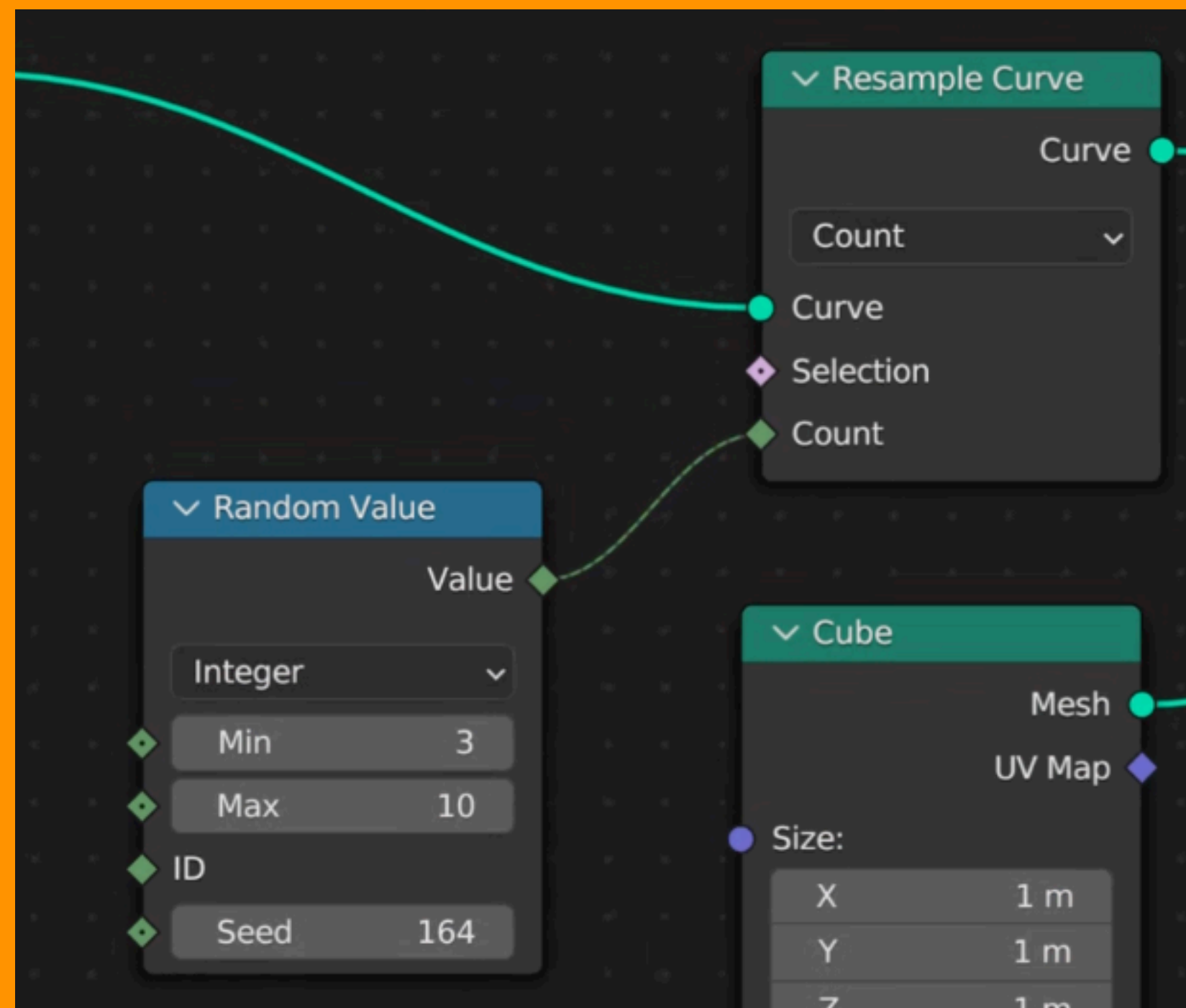
15. Change **Seed** in Random Value



Randomize the amount of cubes

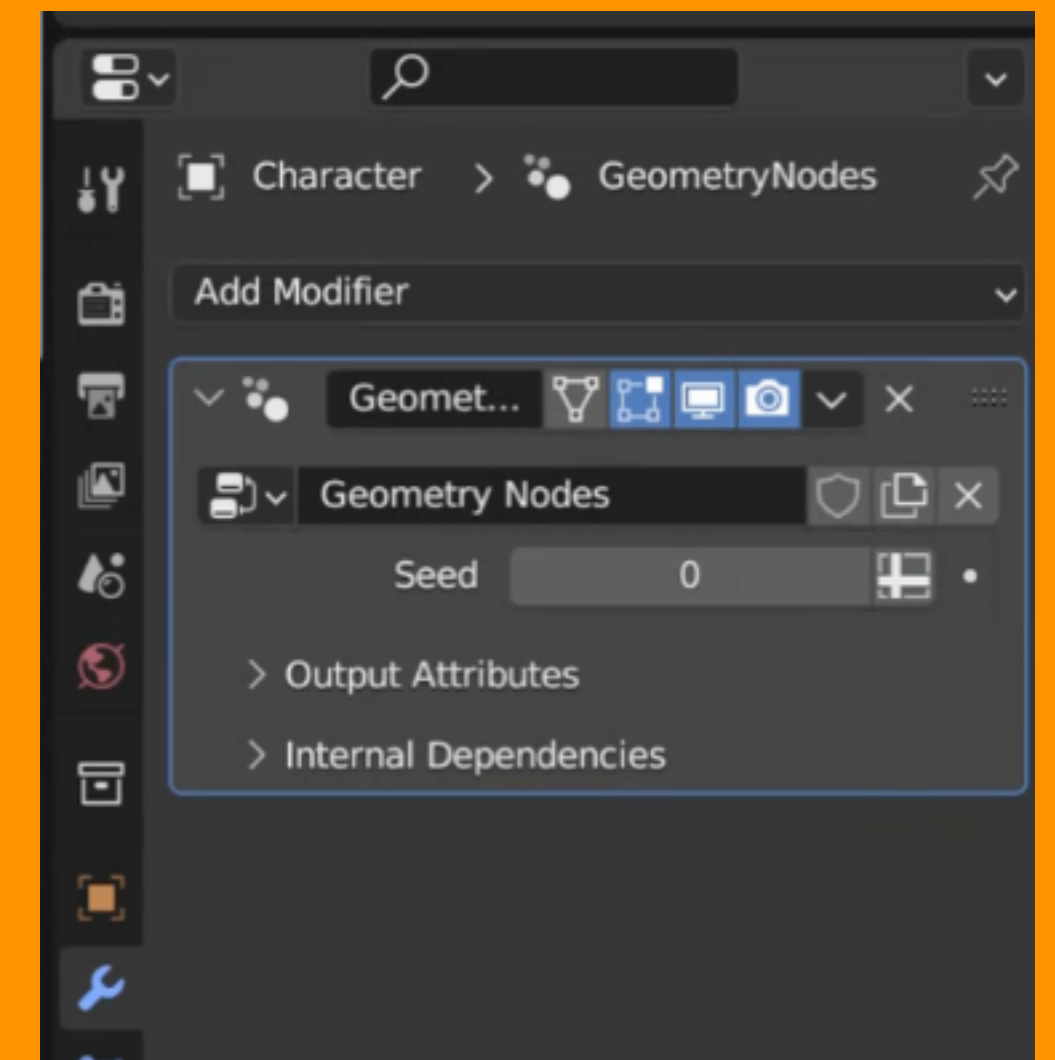
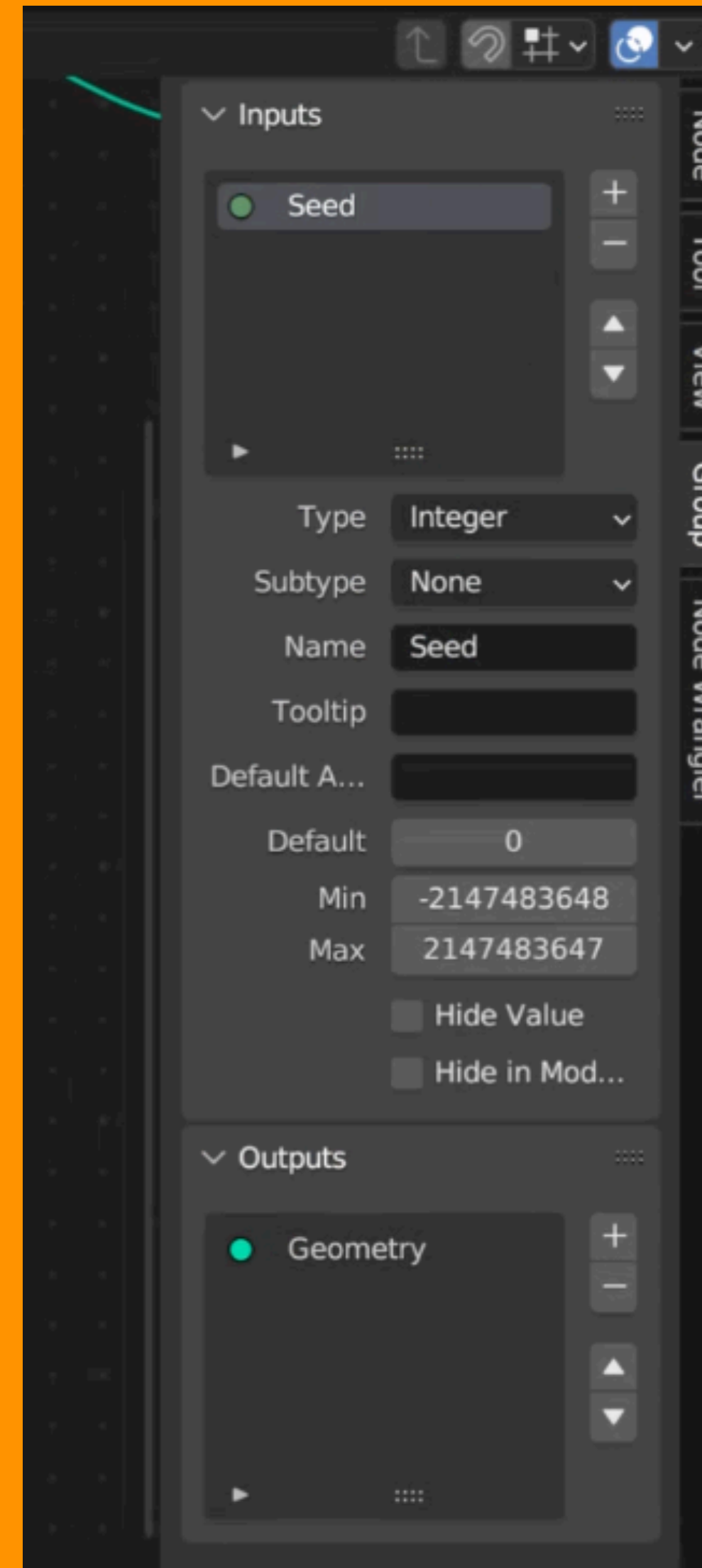
16. Add **Random Value** node

17. Connect to **Count** in Resample Curve



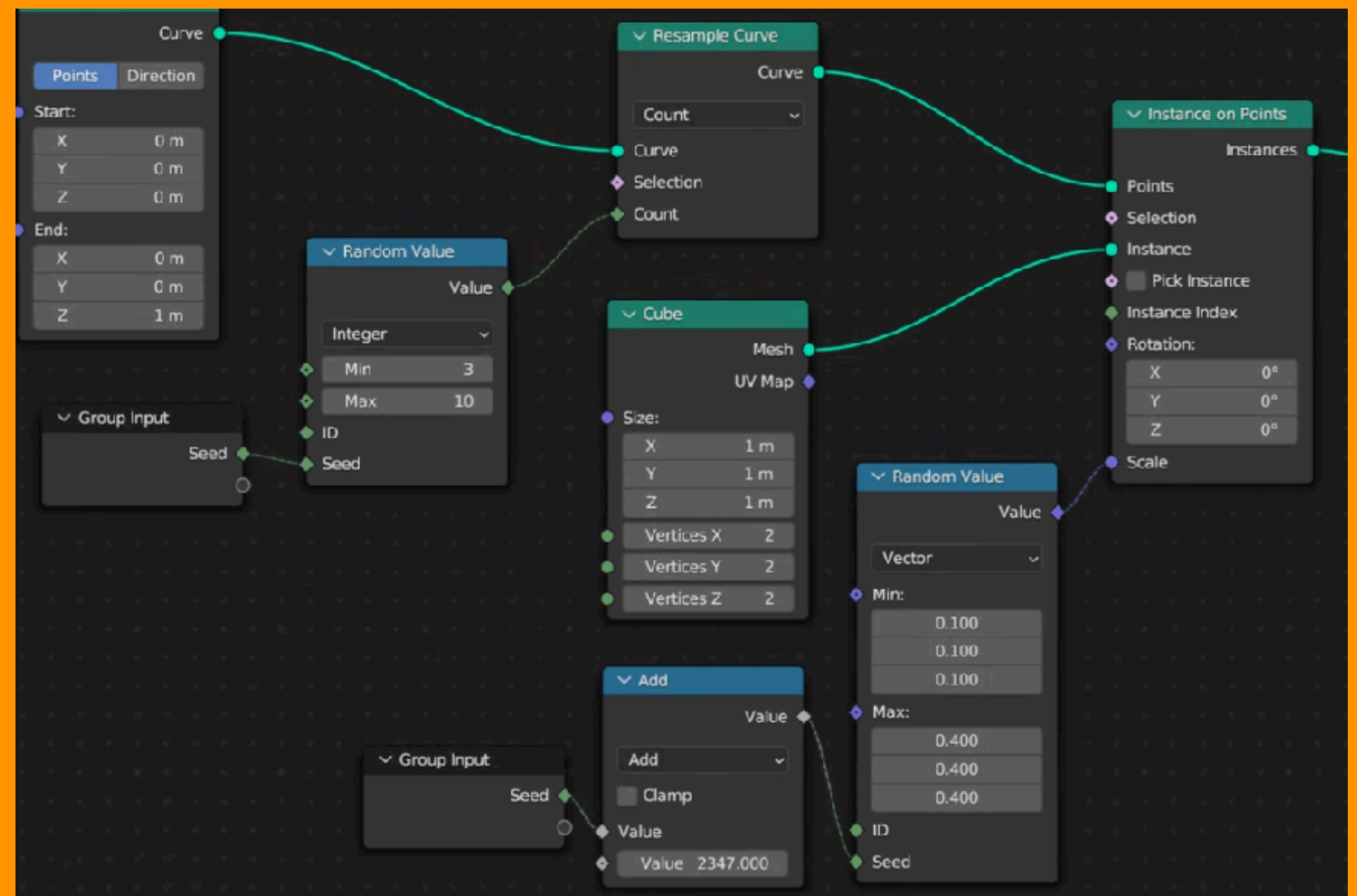
Add Control seeds

18. Add **Input** in **Group** tab
19. Set type to **Integer** and name to **Seed**
20. Remove **Geometry** input
21. 'Seed' setting is now visible in modifier tab, but we're not using it yet...



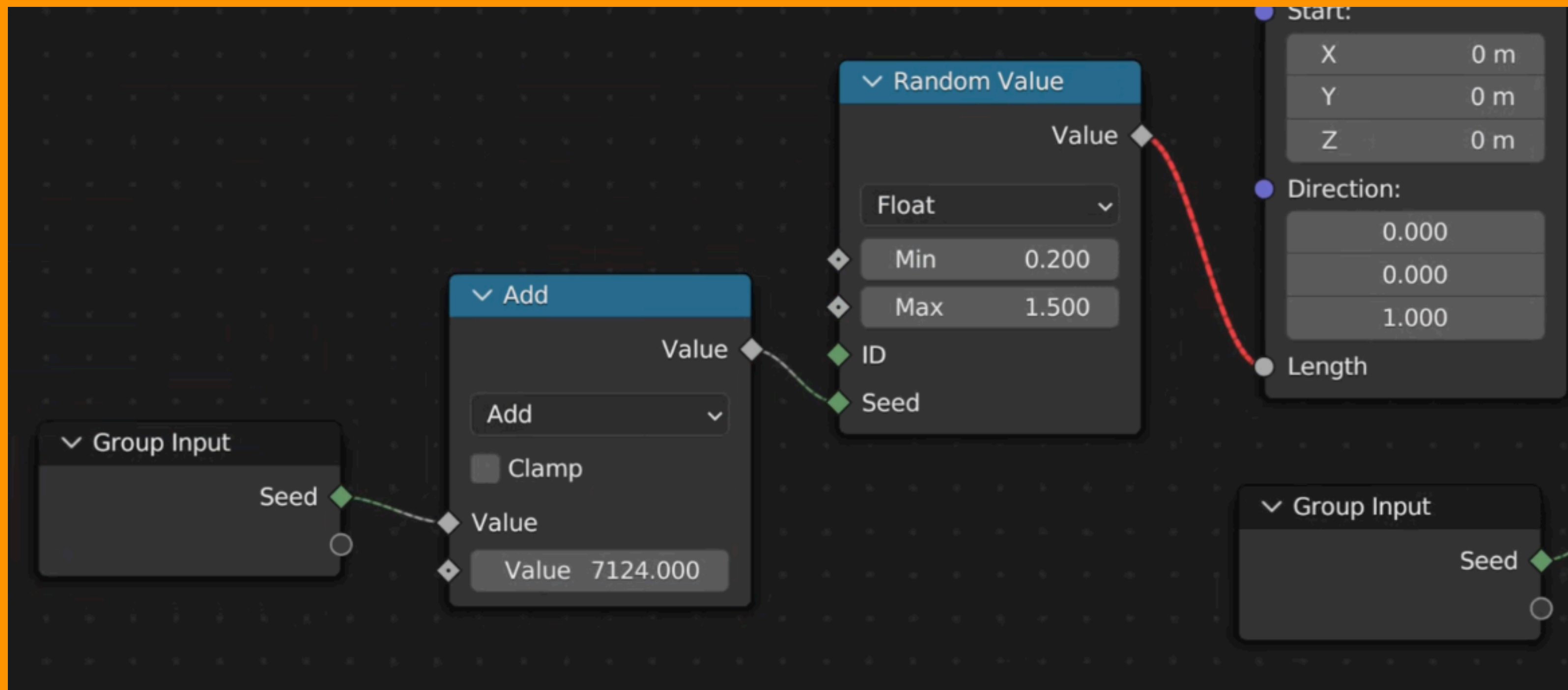
Using Group Input to seed random values

22. Add **Group Input** node
23. Connect **Seed** to Seed in **Random Value (Count)**
24. Duplicate **Group Input** node (shift-d)
25. Add **Math: Add** node and set the **Value** to a random number
26. Connect **Group Input Seed** to **Value**
27. Connect **Add Value** to Seed in **Random Value (Scale)**



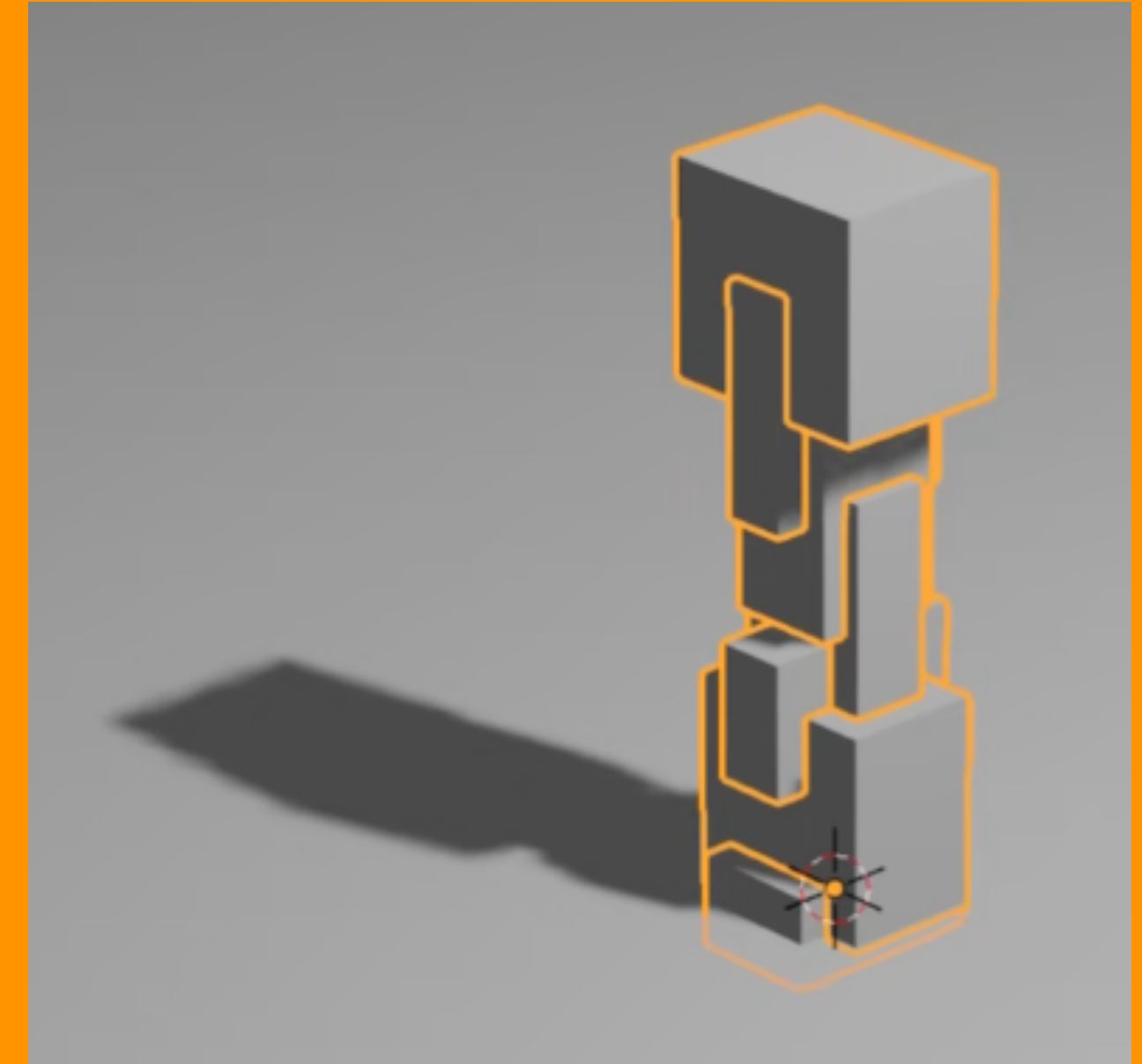
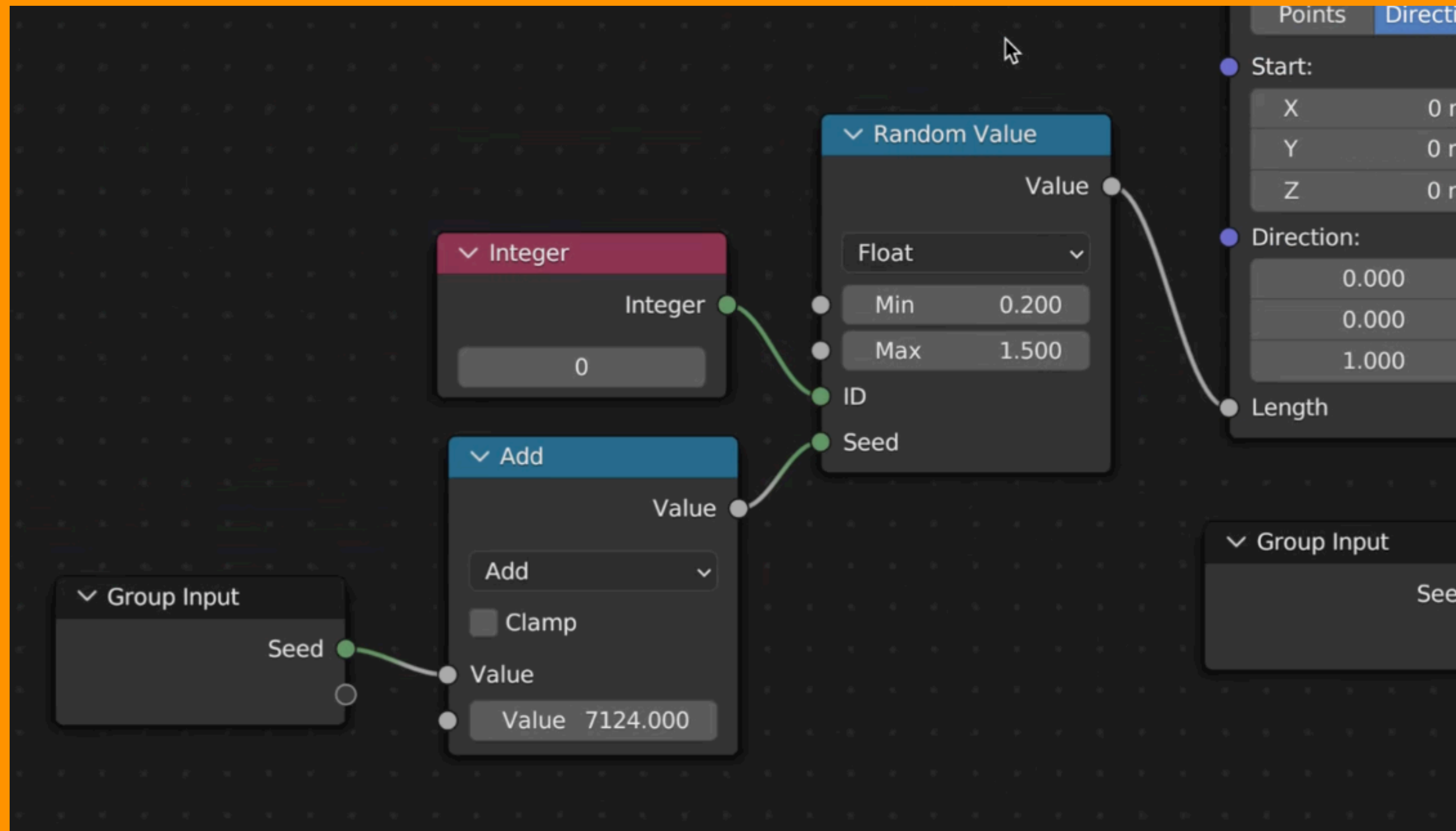
Adding variable height

28. Set **Curve Line** node to **Direction**
29. Add **Group Input**, **Add** and **Random Value** (min is minimum length, max is maximum length)
30. Connect **Random Value** to **Curve Line Length** -> this doesn't work!



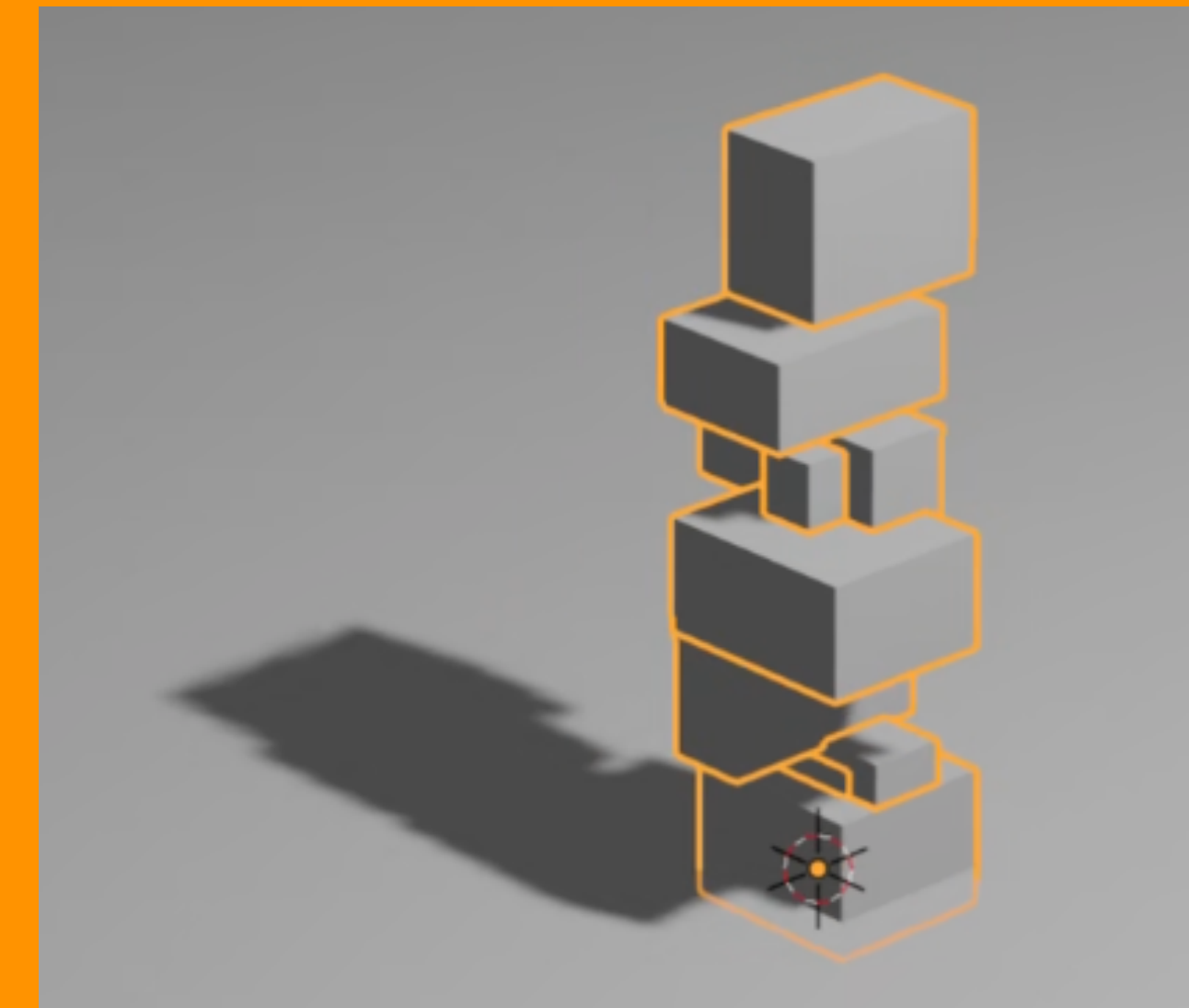
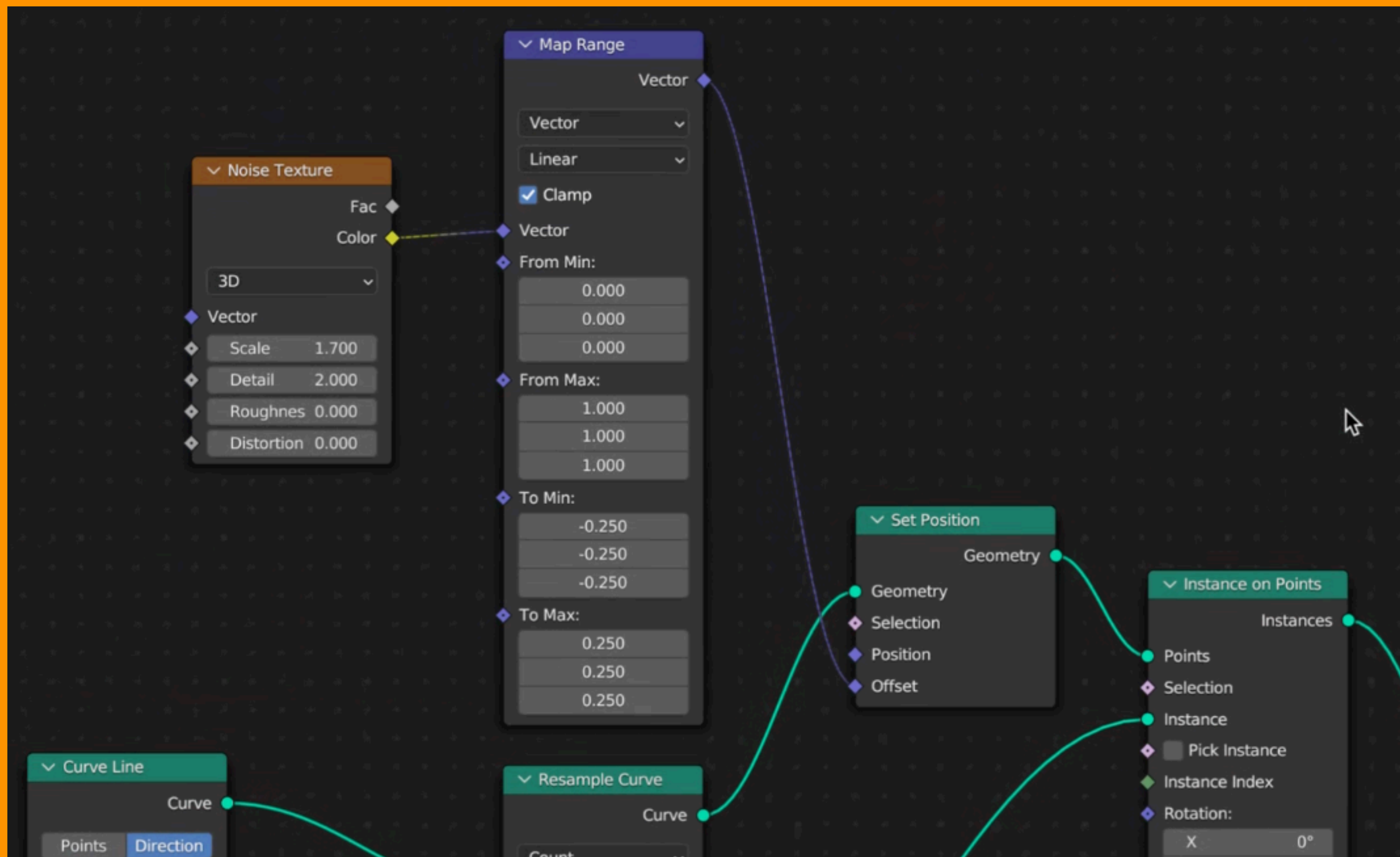
Adding variable height

- 31. Add **Integer** and connect to **ID** in Random Value
- 32. Changing the Seed input parameter now changes the length



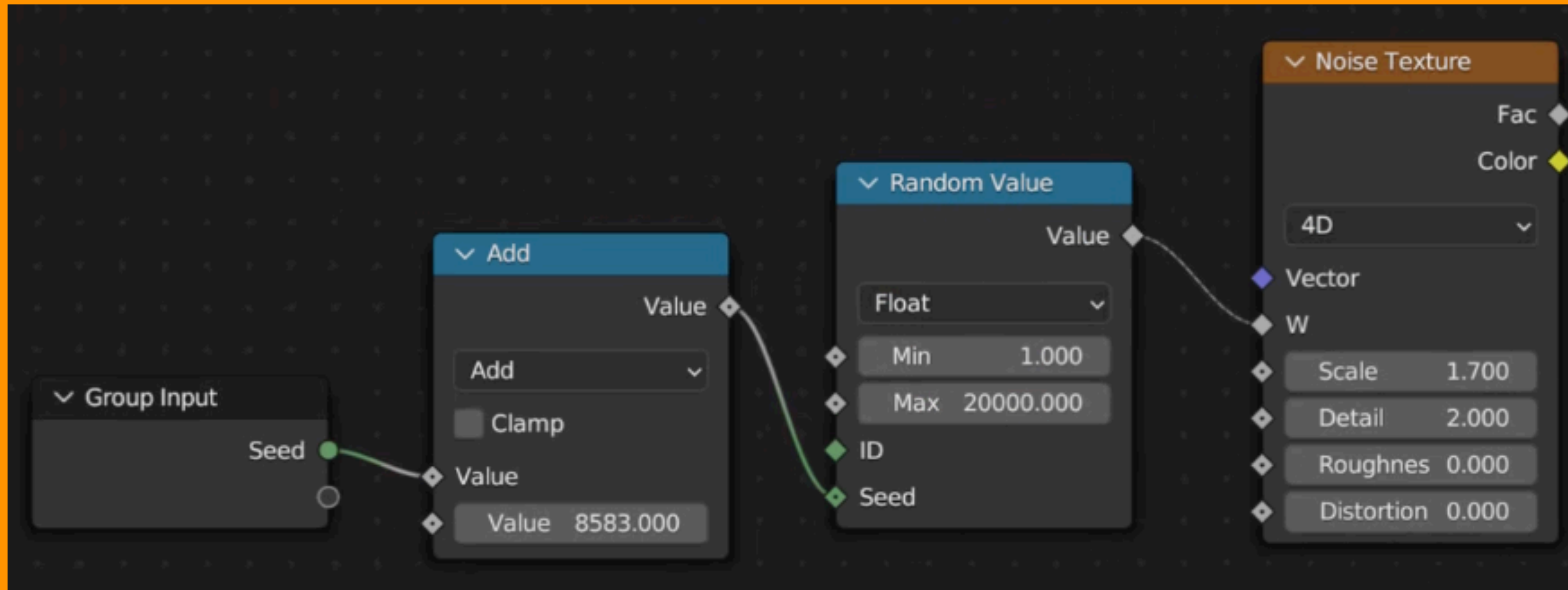
Adding variable form

33. Add **Set Position** node after Resample Curve
34. Add **Noise Texture** and connect Color to new **Map Range**
35. Set Map Range values **To Min: -0.25** and **To Max: 0.25**
36. Connect Map Range to **Set Position Offset**



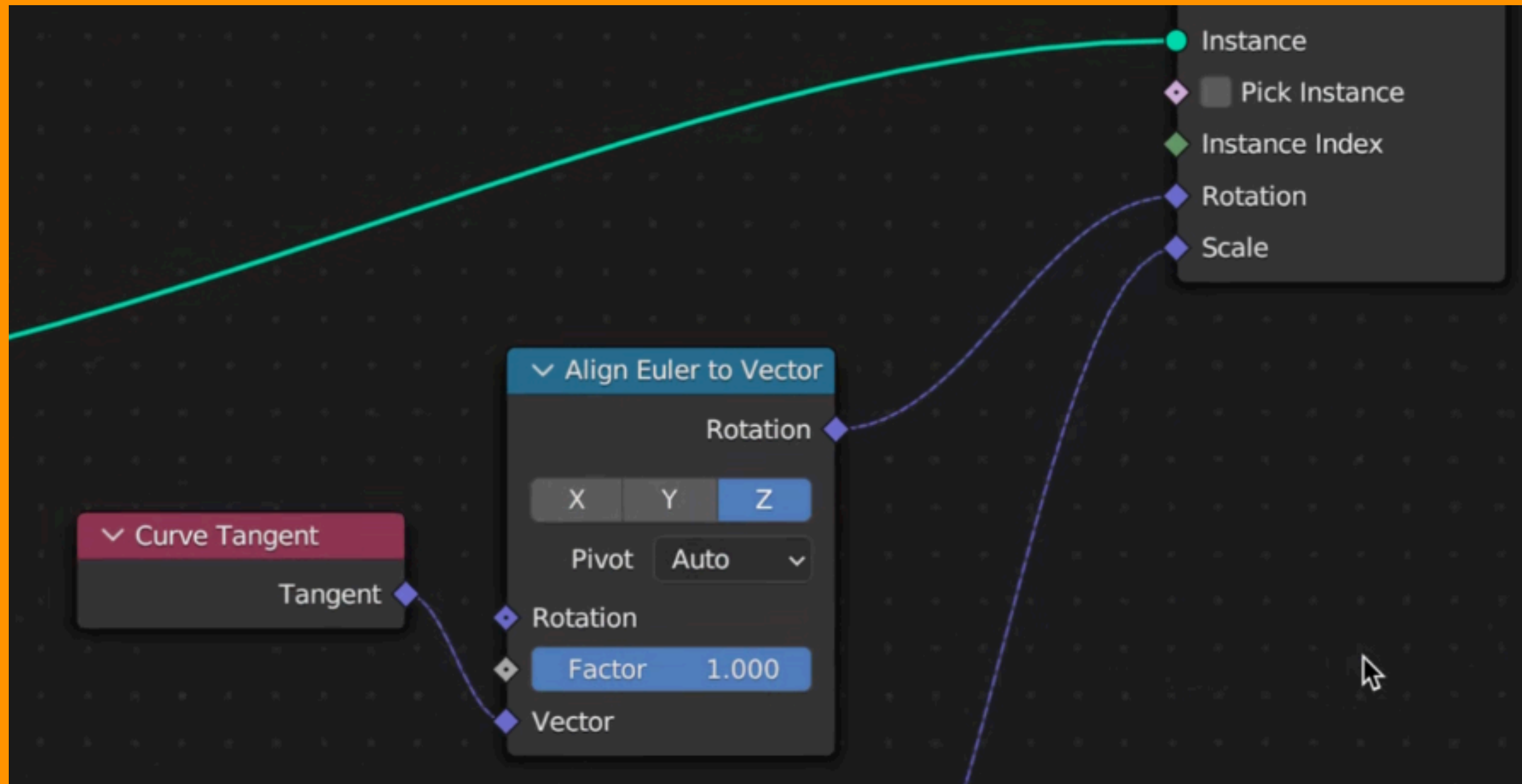
Adding variable form

37. Add **Group Input**, **Add** and **Random Value** (set min to 1, max to 20000)
38. Change Noise Texture dropdown to **4D**
39. Connect **Random Value** to **Noise Texture W** input



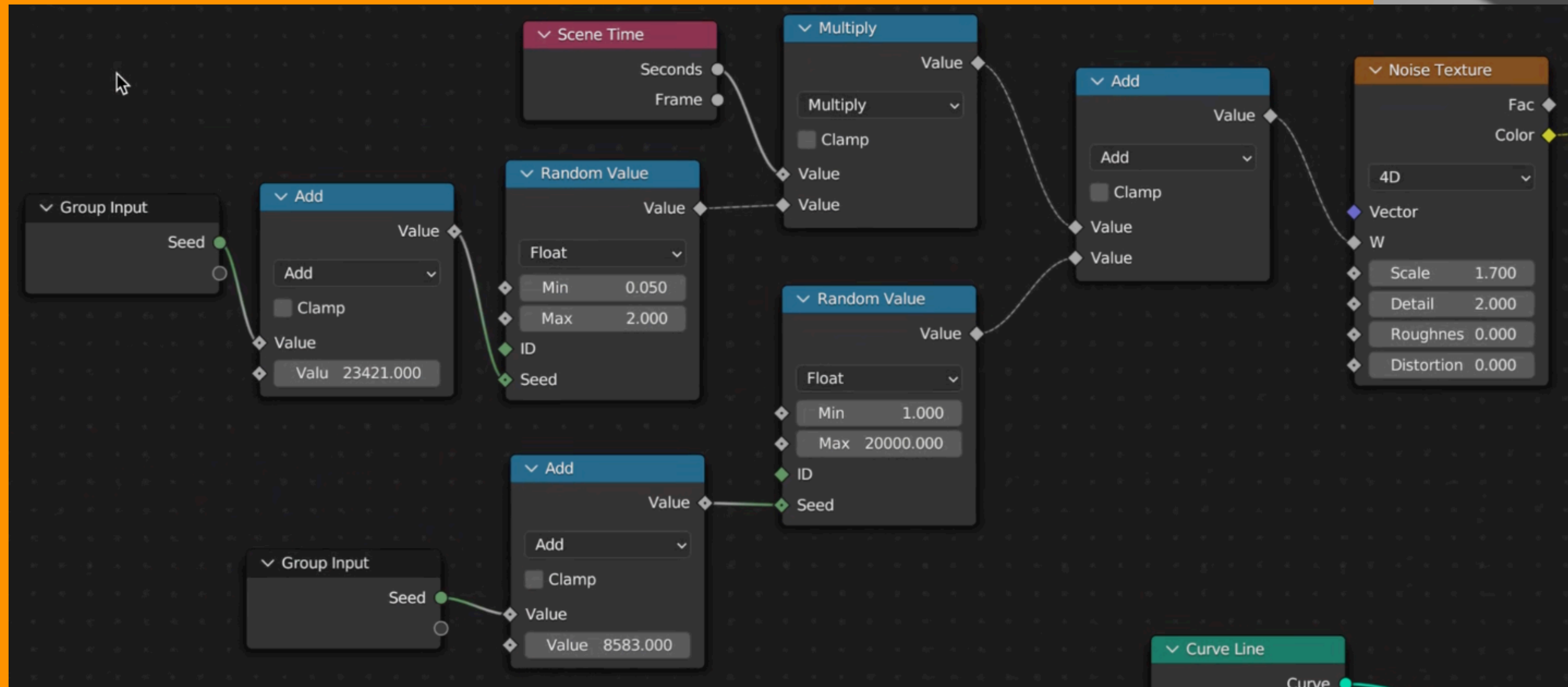
Align cubes to curve direction

- 40. Add **Align Rotation to Vector** node
- 41. Set Align Rotation to Vector to **Z**
- 42. Connect **Rotation** to Rotation in **Instance on Points**
- 43. Add **Curve Tangent** and connect to **Vector** in Align Rotation to Vector



Vibing skyscrapers

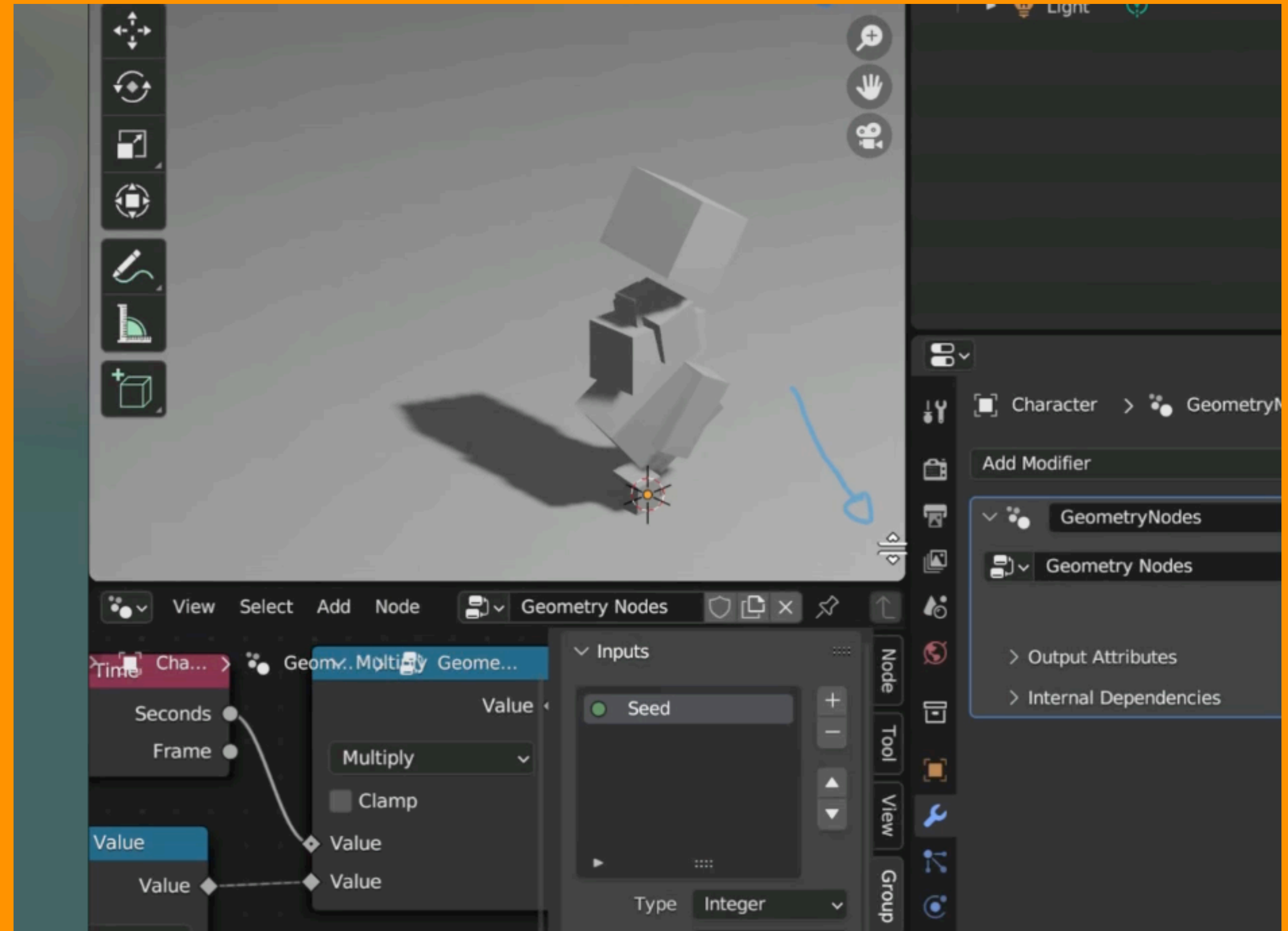
- 44. Add **Add** between **Random Value** and **Noise Texture**
- 45. Add **Scene Time** and add **Math: Multiply** to **Seconds**
- 46. Add **Group Input**, **Add** (with random Value), and **Random Value (Min: 0.050 and Max: 2000)**, connect to **Multiply Value**



Day 1

Showing the timeline

47. Drag new panel, set to **Timeline**



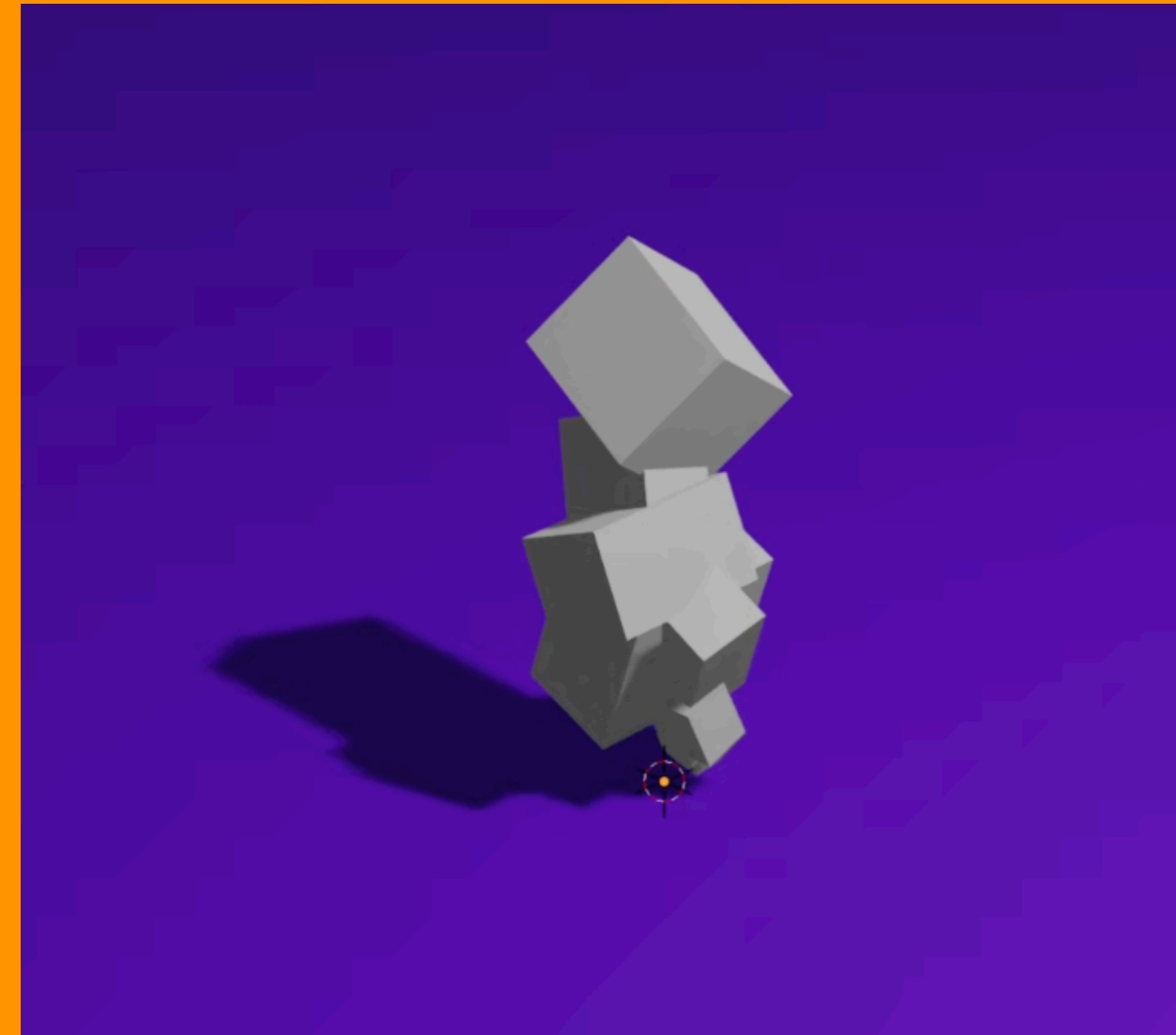
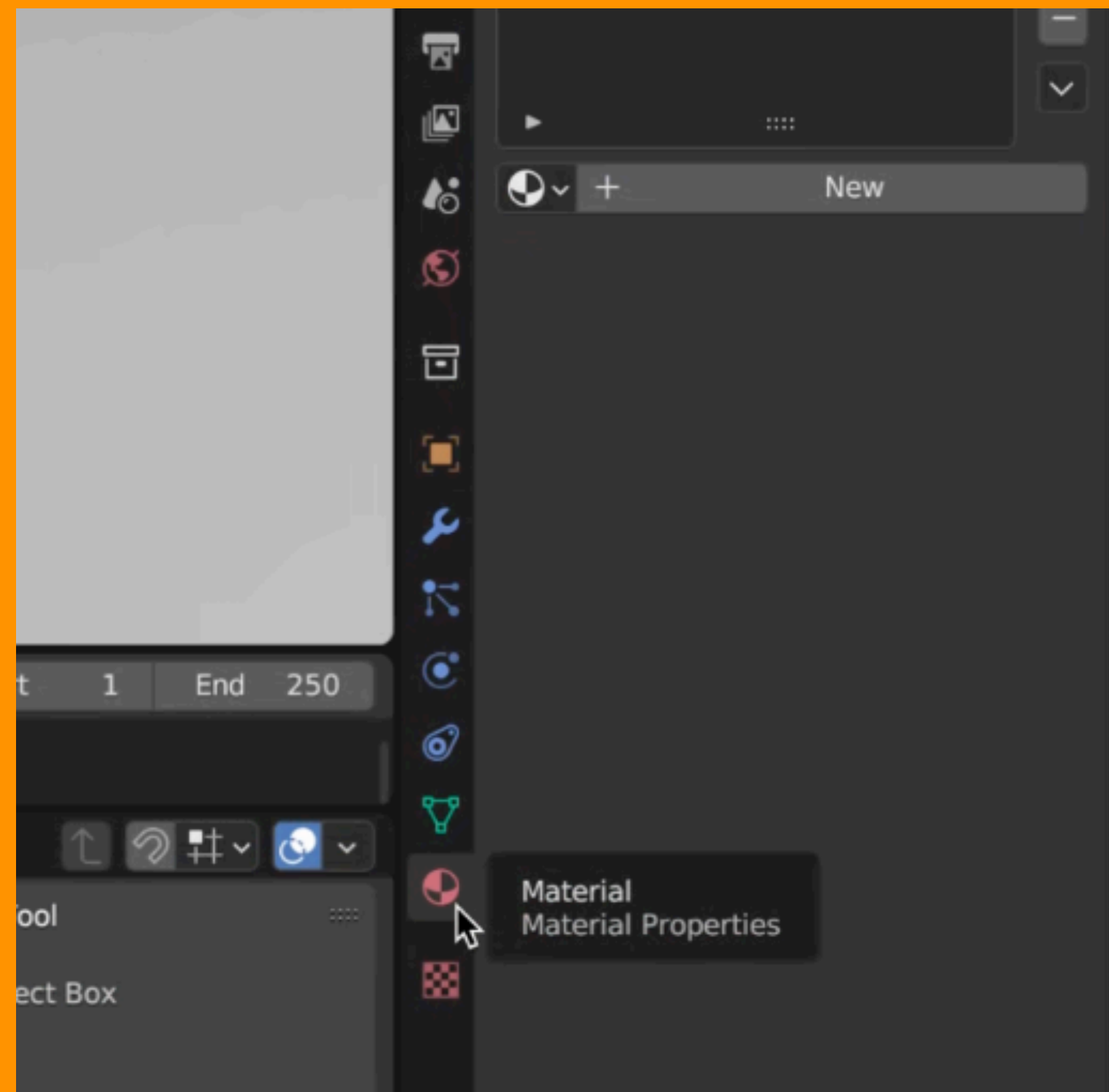
Adding color to the background

48. Select **Ground Plane**

49. Select **Material Properties**

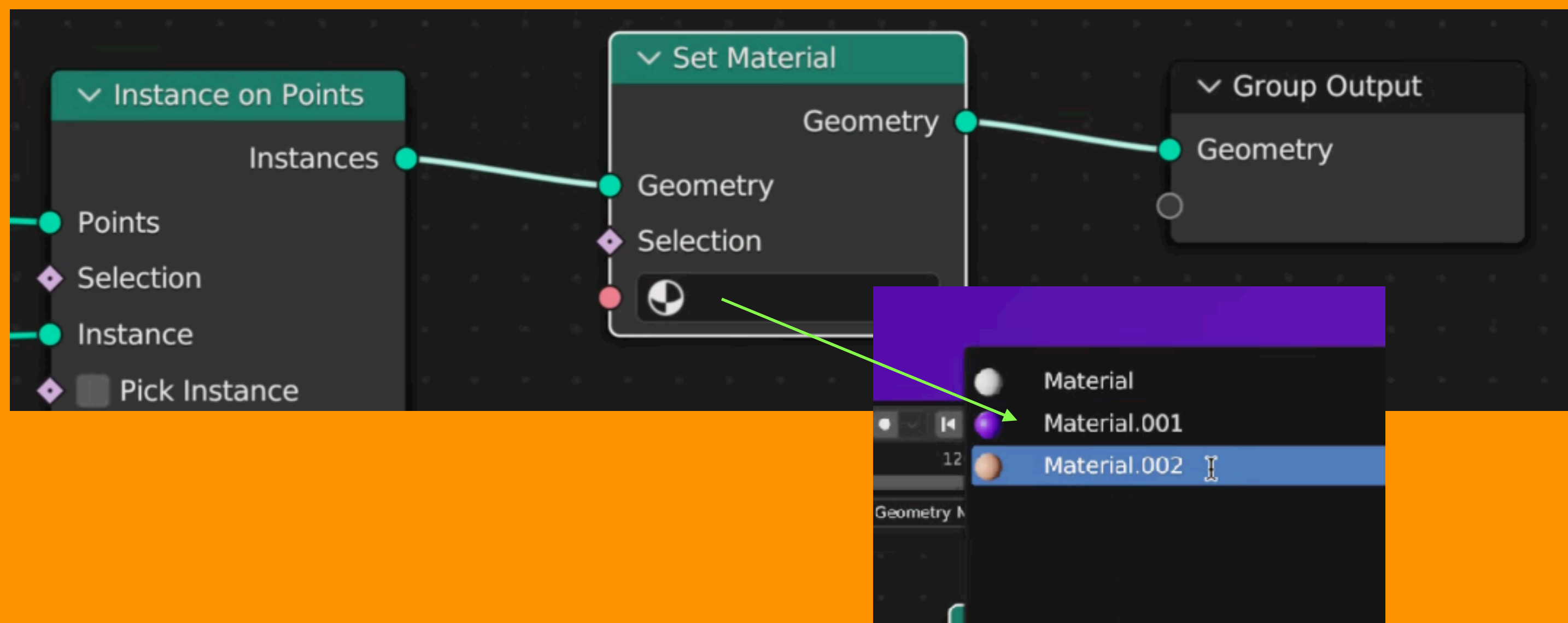
50. Click **New** to create a new material

51. Click **Base Color** to pick a color



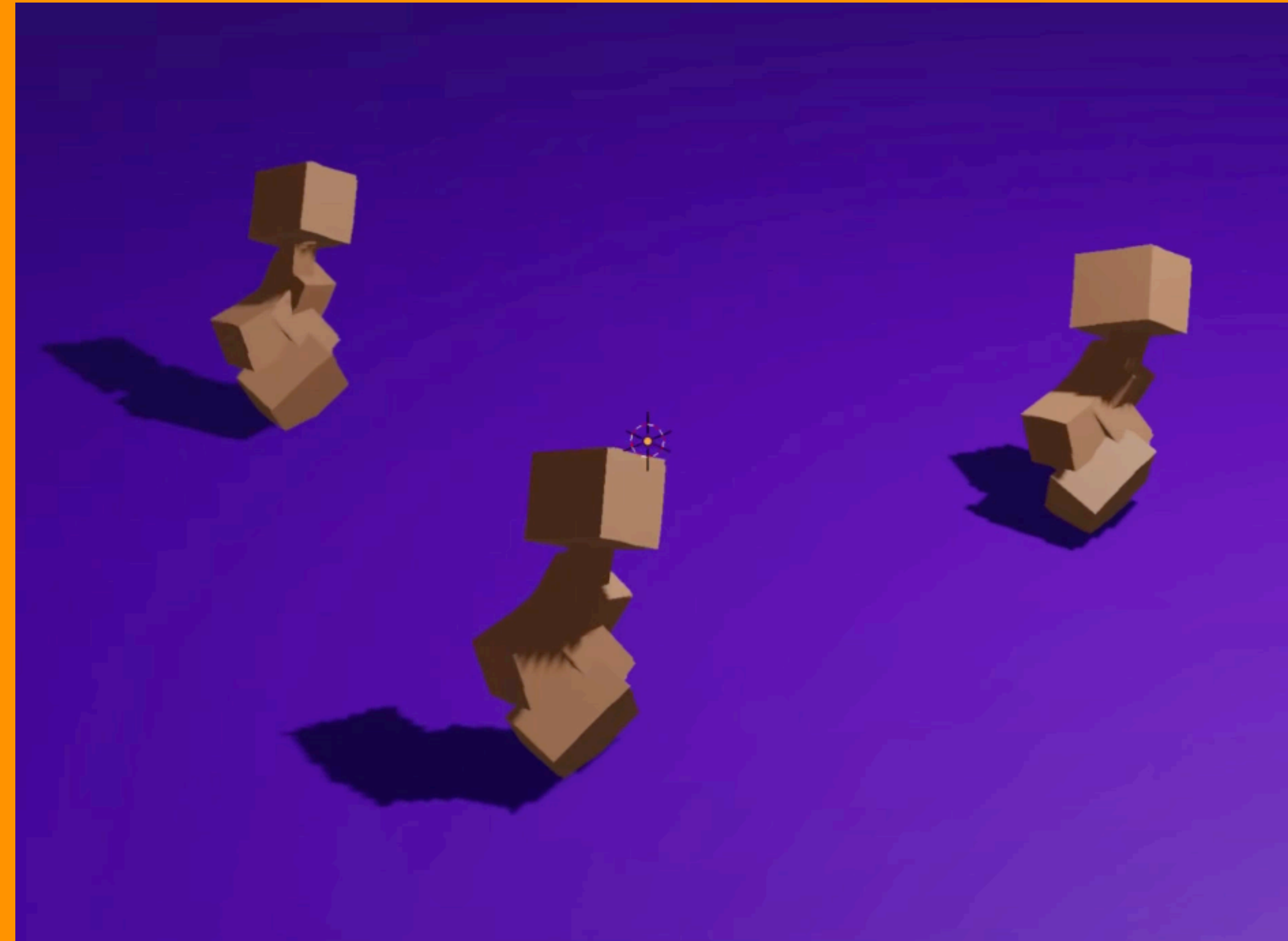
Adding color to your skyscraper

- 52. Click on the **Skyscraper**
- 53. Select **Material Properties**
- 54. Click **New** to create a new material
- 55. Click **Base Color** to pick a color -> nothing happens!
- 56. Add **Set Material** node between Instance on Points and Group Output
- 57. Click on the Material input field en select a material from the list



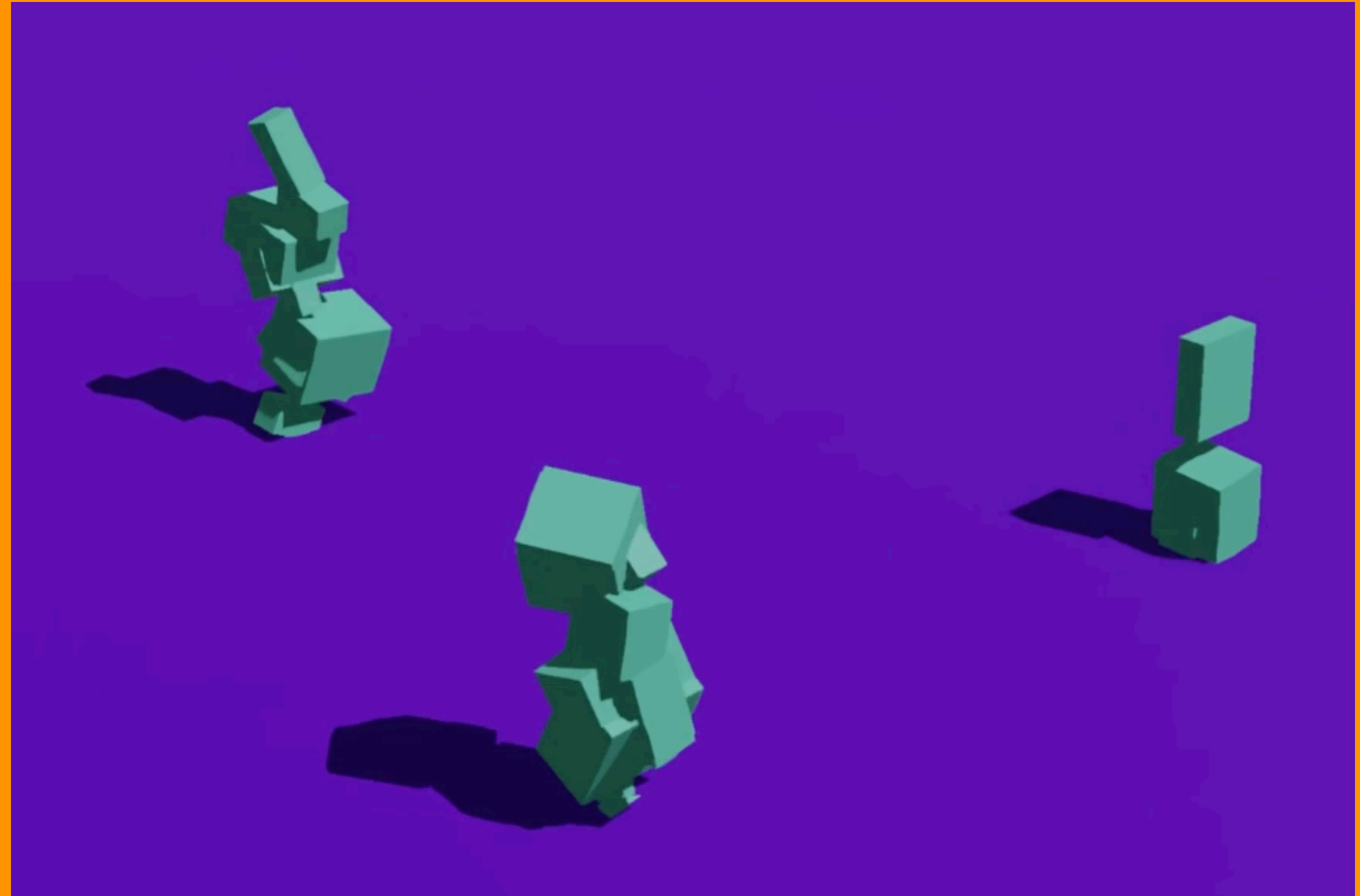
Creating multiple skyscrapers

- 58. **Click** your skyscraper in the viewport
- 59. Hit **Shift-D** and then **Shift-Z**
- 60. **Move** your mouse to find a good spot and **click** to set position
- 61. Do this again to create as many characters as you want!



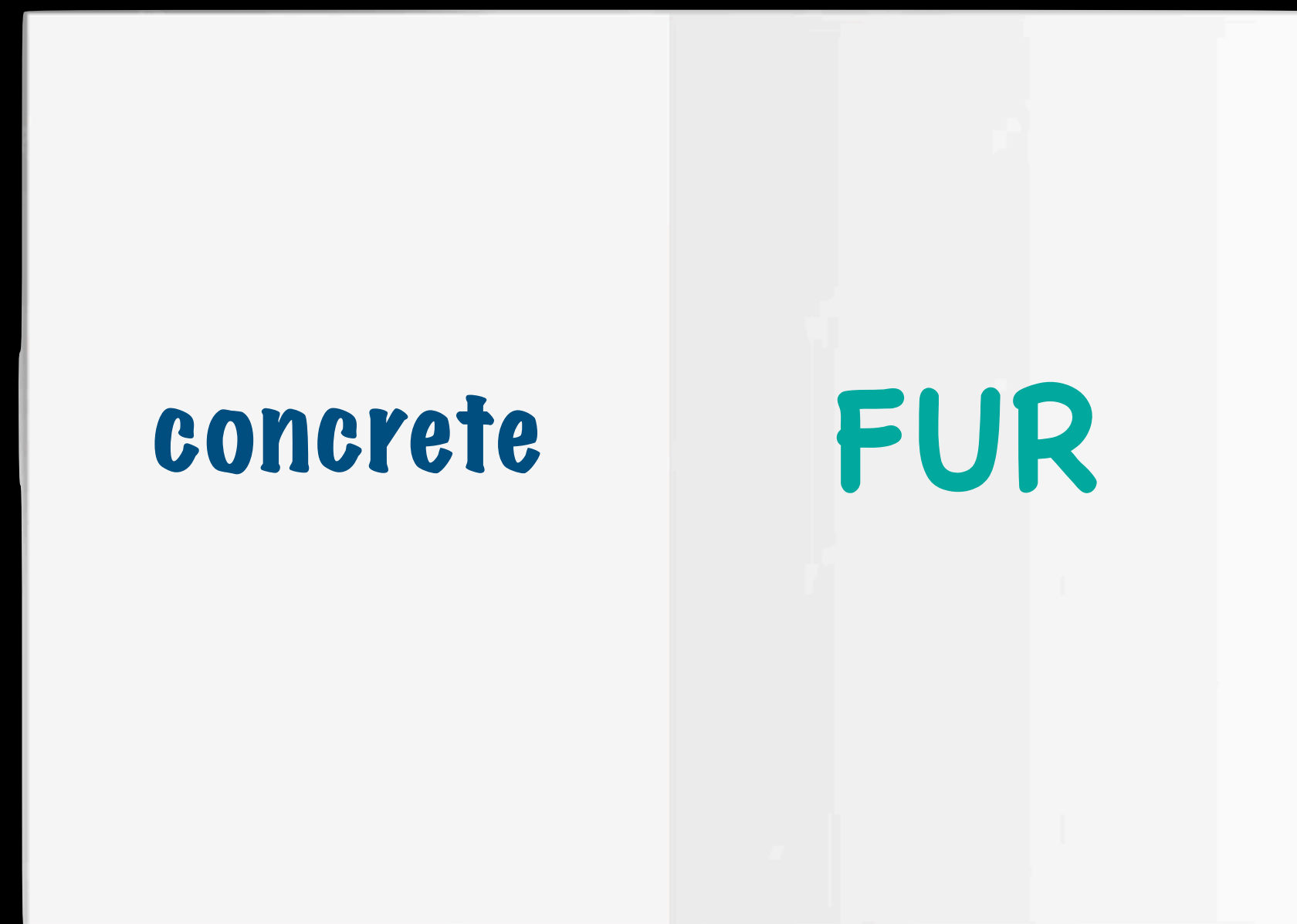
Changing the skyscraper behaviour

- 62. Click on a **skyscraper**
- 63. Click the **Modifier Properties**
- 64. Change the **Seed** parameter



Pairing up

Each pair comes up with two words (representing shape, material, physics, state, etc.) that describe the architectural and natural aspect of their invisible city. Write the two words down on paper.



Task 1

Write a 'procedural premise' — a 3-sentence recipe for a city



Example:

- ▶ The city consists of mirrors
- ▶ Mirrors grow like moss
- ▶ Everything floats; the city has no orientation

Task 2

Define components — architectural vs. natural environment



Example:

- ▶ Mirrors grow like moss
 - ▶ Mirrors are shaped like hills
 - ▶ Hills extend until the horizon
 - ▶ A grid of faces embeds the shape of a mirror